

muchísimocks: An emulated cosmological simulation suite for simulation-based inference

Kate Storey-Fisher,^{1,2*} Raúl Angulo,^{2,3} Marcos Pellejero-Ibañez,⁴ and Drew Jamieson⁵ [ksf: auth order?](#)

¹Kavli Institute for Particle Astrophysics and Cosmology, Stanford University, 452 Lomita Mall, Stanford, CA 94305, USA

²Donostia International Physics Center, Paseo Manuel de Lardizabal 4, 20018 Donostia-San Sebastian, Spain

³IKERBASQUE, Basque Foundation for Science, E-48013, Bilbao, Spain

⁴Institute for Astronomy, University of Edinburgh, Royal Observatory, Blackford Hill, Edinburgh, EH9 3HJ, UK

⁵Max-Planck-Institut für Astrophysik, Karl-Schwarzschild-Straße 1, 85748 Garching, Germany

Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

ksf: NOTE TO COAUTHORS: please define a macro with your initials and favorite color and comment in tex, do not use overleaf comment feature!

Galaxy clustering contains rich cosmological information down to nonlinear scales; however, methods to extract this information are currently limited the high cost of cosmological N -body simulations and by the fidelity of galaxy bias modeling. We develop a new approach to generating approximate yet robust galaxy mocks: we employ the map2map full-field emulator to generate particle displacement fields, and the hybrid Lagrangian bias expansion approach to flexibly model biased tracers. We create a suite of $(1\ h^{-1}\text{Gpc})^3$ second-order bias fields at $z = 0$ spanning 10,000 different cosmologies, and weight these with multiple bias and noise parameter sets, resulting in 320,000 mock 3D density fields. We train a simulation-based inference (SBI) model on the galaxy–galaxy power spectrum $P_{gg}(k)$, the galaxy–matter power spectrum $P_{gm}(k)$, and the galaxy bispectrum $B_{ggg}(k_1, k_2, k_3)$. To test generalizability, we apply it to an independent N -body simulation populated with LRG-like galaxies using extended SubHalo Abundance Matching, and achieve unbiased cosmological constraints. Our results show that $P_{gm}(k)$ is critical for breaking parameter degeneracies, and that including $B_{ggg}(k_1, k_2, k_3)$ improves precision, on Ω_m by 12%, σ_8 by 11% and b_1 by 70%. **ksf: note: if we use only top model instead of ensemble, it jumps to 39% for sigma8, which is significantly different. let's discuss using the top-3 ensemble vs just top model** In convergence tests, we find that 20,000 mocks are sufficient for $P_{gg}(k)$, beyond 320,000 are necessary for $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$, and that additional bias parameter sets provide similar gain as additional cosmologies. These findings are useful for guiding future SBI approaches for galaxy clustering, and demonstrate a promising path towards optimal inference and many other potential applications. The muchísimocks suite is available at [thisurl](#). **ksf: right around 250-word limit**

Key words: methods: numerical – methods: statistical – galaxies: statistics – cosmology: large-scale structure of Universe – cosmological parameters

1 INTRODUCTION

The clustering of galaxies encodes information about the cosmological model describing the fundamental components of the universe. Surveys such as the Sloan Digital Sky Survey (York et al. 2000) and most recently the Dark Energy Spectroscopic Instrument (DESI, DESI Collaboration et al. 2023) have spectroscopically observed millions of galaxies, producing detailed maps of a large volume of the observable universe. Current and upcoming surveys such as Euclid (Euclid Collaboration et al. 2025) and the Roman Observatory (Spergel et al. 2015; Wang et al. 2022) will continue to deliver competitive and complementary galaxy clustering data sets.

Standard approaches to analyzing this spectroscopic survey data rely on analytic techniques to model summary statistics of the galaxy field, and Monte Carlo methods to sample the parameters of interest

(e.g. Alam et al. 2017; Philcox & Ivanov 2022). Analyses are thus largely restricted to two-point statistics on quasi-linear scales, which contain robust cosmological information but do not fully capture the information content in the data. For instance, the DESI analyses use a perturbation theory approach, and their analysis is limited to $k_{\text{max}} = 0.2$ for the power spectrum and $k_{\text{max}} = 0.12$ for the bispectrum (Adame et al. 2025; Novell-Masot et al. 2025). Further, as surveys become more sensitive and statistical uncertainties improve, systematic errors will become increasingly relevant, and the effect of various systematics on clustering statistics—both two-point and higher-order statistics—is difficult to model.

A new approach known as simulation-based inference (SBI) has shown promise for cosmological inference from galaxy clustering, as it obviates these limitations. In the SBI approach, only a forward model of the data is required, and the posterior of the input parameters is approximated conditioned on the observed data. In the case of galaxy clustering, one can thus forward model the full galaxy

* E-mail: kstoreyf@stanford.edu

field and directly measure any statistic, without the need for analytic modeling. Similarly, one can directly forward model the effects of systematics on the galaxy field, and the statistics can be simply measured without the need for additional modeling specific to each statistic. Finally, Monte Carlo methods typically assume a Gaussian form for the likelihood (and must estimate a nontrivial covariance matrix), even though higher-order statistics in particular are known to have non-Gaussian properties (Hahn et al. 2018); SBI makes no assumptions about the form of the likelihood. Further, in the case of galaxy clustering one can learn optimized statistics of the galaxy field for the parameters of interest, effectively using the full information in the galaxy field.

While promising, SBI faces challenges for galaxy clustering analyses. One key issue is that SBI typically requires many simulations, but cosmological simulations are very computationally expensive. An N -body simulation with the volume and resolution to accurately describe galaxy clustering into the nonlinear regime costs at least hundreds (and up to millions) of CPU hours, and thousands to hundreds of thousands of simulations are generally needed for SBI. Another challenge is modeling how galaxies are connected to the underlying dark matter distribution, which involves complex and not yet well-understood galaxy formation physics; to be useful for SBI, then, models must be relatively fast, flexible enough to model various galaxy populations, and accurate to as small scales as possible.

There has been significant recent work that has demonstrated SBI as a promising path forward for the field. The SimBIG collaboration employs a forward model based on the $2,000 (1h^{-1}\text{Gpc})^3$ simulations of the Quijote Latin hypercube (Villaescusa-Navarro et al. 2020); they use the halo occupation distribution (HOD) model to populate the simulations with galaxies, choosing 10 HODs per cosmology for a total of 20,000 simulations (Hahn et al. 2023c). Using the power spectrum and applying to SDSS BOSS data (Dawson et al. 2016), they achieved constraints on σ_8 that are 27% more precise than the standard analysis thanks to the information in small scales (Hahn et al. 2022); subsequent analyses have applied SimBIG to the bispectrum (Hahn et al. 2023a), the marked power spectrum (Massara et al. 2024), and the full galaxy field (Lemos et al. 2023; Hahn et al. 2023b). Tucci & Schmidt (2024) take the effective field theory of large-scale structure (EFTofLSS) approach, generating 10^5 simulations and employing neural posterior estimation using the power spectrum and bispectrum out to $k_{\text{max}} = 0.2$; as this model is fully perturbative, they are able to consider a Euclid-size volume of $63 (h^{-1}\text{Gpc})^3$. By performing a direct comparison to standard inference, they find that the off-diagonal elements of the covariance matrix are important for constraining σ_8 accurately, while allowing for a non-Gaussian likelihood is less important. Using the same forward model, Nguyen et al. (2024) show that full-field inference yields over a factor of five improvement on the constraint on σ_8 compared to the power spectrum and bispectrum at the same scale cut of $k_{\text{max}} = 0.12$. Recent work has also explored forward models based on more physically motivated models of galaxy formation, albeit at the expense of much smaller volumes. de Santi et al. (2023) employ the 1,000-simulation Latin hypercubes of the CAMELS suite (Villaescusa-Navarro et al. 2021), which are hydrodynamical simulations with $(25 h^{-1}\text{Mpc})^3$ volumes, to perform SBI using graph neural networks; they find that their inference on Ω_m is robust to galaxy formation physics.

Analyses that rely on N -body simulations, as some of these SBI approaches do, are limited by the high computational cost of these simulations. Significant effort has been spent developing *emulators* that learn the mapping from cosmological parameters to statistics of the density field, trained on a relatively small set of cosmologies and then able to effectively interpolate between them. Recently,

full-field emulators have been developed that go beyond summary statistics, instead learning the full particle displacement field. The emulator *map2map* (Jamieson et al. 2023) uses a convolutional neural network-based architecture to map from Zel’dovich approximation displacements to N -body displacements, trained on the Quijote simulation suite (Villaescusa-Navarro et al. 2020). It has been used in a range of applications, such as producing large-volume emulated simulations (Scoggins et al. 2025); variants of *map2map* have been designed to emulate modified gravity (Saadeh et al. 2025) and to learn the mapping from nonlinear structures to the initial conditions of the universe (Doerer et al. 2024). There has been continued work on developing full-field emulators, with a range of training data and neural architectures (Kang et al. 2026; Ullmo et al. 2026; Jacobus et al. 2026).

Robustly modeling the galaxy–dark matter connection is a key challenge (Wechsler & Tinker 2018). Halo-based models such as the halo occupation distribution (e.g. Peacock & Smith 2000; Seljak 2000; Zheng et al. 2005) populate dark matter halos with galaxies based on halo mass and properties, but their flexibility degrades their robustness, particularly for higher-order statistics (e.g. Contreras et al. 2024; Wang et al. 2026). More physically motivated models like subhalo-abundance matching (e.g. Vale & Ostriker 2004; Shankar et al. 2006; Conroy et al. 2006) are more robust, but require merger trees which are more computationally expensive; even more complicated models like semi-analytic models (e.g. Somerville et al. 2008; Henriques et al. 2015; Stevens et al. 2016; Somerville & Davé 2015) and hydrodynamical simulations (e.g. Springel et al. 2005; Vogelsberger et al. 2014; Schaye et al. 2015) are both expensive and make specific choices regarding galaxy formation, which limits their generalizability. EFTofLSS offers a perturbative approach that is robust on large scales, but breaks down on small scales (Bernardeau et al. 2002; Desjacques et al. 2017).

The hybrid Lagrangian bias expansion (or hybrid effective field theory) offers an approach that is accurate into the small-scale regime while remaining computationally relatively low-cost (Modi et al. 2020; Kokron et al. 2021; Zennaro et al. 2023). It combines the Lagrangian bias expansion, which models the galaxy field as a summation of operators of the dark matter field, with particle displacement and velocity fields from N -body simulations to advect them to Eulerian space. This approach has been shown to be effective down to $k \sim 0.6$, and has been used to construct accurate emulators of the power spectrum (Pellejero-Ibanez et al. 2022, 2023b). Pellejero Ibanez et al. (2023a) showed that emulated N -body displacements with *map2map* can be used as inputs to the hybrid bias expansion to provide an accurate model of galaxy abundance. They found 1–2% precision on the power spectrum down to $k \sim 0.6$; while there were some biases at intermediate scales, these were absorbed by nuisance parameters describing stochasticity. Notably, they found that the robustness held with variations in the cosmological parameters, demonstrating promise for using this approach for cosmological inference.

In this paper, we construct a suite of three-dimensional mock galaxy fields using our forward model based on the hybrid Lagrangian bias expansion with emulated displacements from the full-field *map2map* emulator. The suite, called *muchísimocks*, consists of both real- and redshift-space mocks at $z = 0$ and spans galaxy bias and cosmological parameter space. This suite opens the door to simulation-based inference for cosmology with a hybrid bias expansion-based galaxy field, as well as with a very large number of simulations given the speed of the emulator. We demonstrate this approach by training a neural posterior inference model on summary statistics computed on *muchísimocks* in real space; in this work we

focus on the galaxy–galaxy power spectrum, the galaxy–matter cross power spectrum, and the galaxy bispectrum. Our maximal training set consists of 10,000 cosmologies and 32 bias parameter sets per cosmology, for a total of 320,000 training mocks; we investigate the dependence of the inference performance on the number of training mocks, with an eye towards guiding the development of future SBI pipelines.

This paper is organized as follows. In §2, we present our forward model for galaxy fields, including details of the N -body emulator, hybrid bias expansion, and noise model. We introduce the methodology used for our simulation-based inference pipeline in §3, including the summary statistics analyzed and the construction of our training set and various test sets. In §4 we show our results on these test sets, as well as analyses of the scale dependence and convergence with size of training data. We discuss the implications of our results in §5, and summarize our findings and conclude in §6.

2 THE MUCHISIMOCKS SUITE

The muchisimocks forward model is based on emulated N -body displacements (§2.1) and the hybrid Lagrangian bias expansion (§2.2), as well as a full-field noise model (§2.3). An overview of the model is given in the upper panel of Figure 1.

2.1 Emulating the displacement field with map2map

The nonlinear matter distribution can be simulated to high accuracy with N -body simulations. However, simulations with the resolution and volume required to analyze current galaxy surveys are very expensive, and training simulation-based inference models requires a very large number of simulations. We thus turn to a field-level emulator of the matter distribution, map2map (Jamieson et al. 2023).

The density field can be approximated at linear order by the Zel’dovich approximation (ZA, also known as first-order Lagrangian perturbation theory, 1LPT), based on a realization of an initial conditions field. The map2map emulator learns a mapping from the ZA field at $z = 0$ to the nonlinear N -body displacement field at that redshift, effectively learning the correction from linear to nonlinear displacements. The emulator is trained on the 2000 simulations of the Quijote simulation suite that span a Latin hypercube (LH) of five cosmological parameters: Ω_c , σ_8 , h , Ω_b , and n_s (Villaescusa-Navarro et al. 2020). Each has a volume of $(1 h^{-1}\text{Gpc})^3$ with 512^3 particles, giving particle mass resolutions in the range 2×10^{11} to $1 \times 10^{12} h^{-1} M_\odot$. map2map also trains a separate network to predict the nonlinear velocity field from the ZA velocities. For this work, we use a modified version of this velocity prediction: following the BACCO model, we assign Lagrangian particles that end up in halos as all having the same velocity, that of the center-of-mass velocity of the parent halo (see Pellejero Ibanez et al. 2023a for details). [ksf: please check this sentence!](#) Once trained, map2map can generate the displacement and velocity fields for any input ZA field, taking a few minutes on a GPU for a $(1 h^{-1}\text{Gpc})^3$ volume. While we do not utilize the velocity fields in this work, as we limit to real-space analysis, these will be important for future analyses including redshift-space distortions as well as other applications involving the velocity distribution; we release them as part of the muchisimocks suite.

In more detail, map2map consists of two convolutional neural networks (CNNs), one to predict the matter density field and the other to predict the velocity field. The particle displacements are structured on a 3D grid as they are functions of their initial Lagrangian space position, so the CNN processes the inputs and outputs as 3-channel

images. As most of the cosmological parameters only affect the input power spectrum that is used to create the ZA field, these are not needed as additional inputs to the model. However, the matter density Ω_m affects the N -body evolution, so is included as an input “style” parameter; this is incorporated via the approach of the StyleGAN2 (Karras et al. 2020), in which the style parameters modulate the convolution kernels. The model uses a V-net-like architecture, with 3 layers of downsampling and upsampling, and leaky ReLU activations. For its loss function, it combines the log mean squared error of the Lagrangian particle displacements and Eulerian density field. We note that there is a newer version of map2map that includes redshift dependence, and predicts the velocity field as a derivative of the position field (Jamieson et al. 2024); however, as we focus on $z = 0$ and also require a modification of the velocity field as described above, we use the original map2map implementation.

For our mock suite, we define a Latin hypercube of 10,000 sets of cosmological parameters: Ω_c , σ_8 , h , Ω_b , and n_s . We use uniform priors for each parameter that span $\sim 10\sigma$ around the Planck Collaboration et al. (2014) values of the cosmological parameters, following Angulo et al. (2021) (with the range for σ_8 extended to slightly lower values); these are shown in Table 1. For each of these cosmologies, we generate a ZA field based on a different random seed, and pass that field through map2map to obtain the nonlinear particle displacements. The map2map-emulated particle displacements are a key ingredient in modeling the galaxy bias fields, as discussed in the next section.

2.2 Modeling tracers with the hybrid bias expansion

To model biased tracers of the matter field, we use the hybrid Lagrangian bias expansion, also known as hybrid effective field theory (HEFT, Modi et al. 2020; Kokron et al. 2021; Zennaro et al. 2023). We first define a mapping from initial (Lagrangian) space coordinates $\mathbf{q}$ to Eulerian space coordinates $\mathbf{x}$ as $\mathbf{x} = \mathbf{q} + \Psi(\mathbf{q})$, where $\Psi(\mathbf{q})$ is the Lagrangian space displacement field. This advection from Lagrangian to Eulerian space effectively expresses the overdensity field as weighted contributions of the initial density field that end up in a given Eulerian location. The displacement term $\Psi(\mathbf{q})$ can be computed with Lagrangian perturbation theory, but this is only accurate down to mildly nonlinear scales; here we take the “hybrid” approach which uses N -body displacements to compute $\Psi(\mathbf{q})$, significantly improving the accuracy down to small scales.

The galaxy overdensity field in Lagrangian space can be described to second order as

$$\delta_g(\mathbf{q}) = b_1 \delta_L(\mathbf{q}) + b_2 \left[\delta_L^2(\mathbf{q}) - \langle \delta_L^2 \rangle \right] + b_{s^2} \left[s^2(\mathbf{q}) - \langle s^2 \rangle \right] + b_\Delta \nabla^2 \delta_L(\mathbf{q}) \quad (1)$$

where $\delta_L(\mathbf{q})$ is the linear matter density in Lagrangian space; $s^2(\mathbf{q})$ is the shear field, defined by the traceless part of the tidal field, given by $s^2(\mathbf{q}) = s_{ij}(\mathbf{q}) s^{ij}(\mathbf{q}) = \delta_i \delta_j \phi(\mathbf{q}) - \frac{1}{3} \delta_{ij}^K \delta_L^K(\mathbf{q})$, where $\phi(\mathbf{q})$ is the linear gravitational potential in Lagrangian space $\nabla^2 \phi(\mathbf{q}) = 4\pi G \bar{\rho} \delta_L(\mathbf{q})$, and δ_{ij}^K is the Kronecker delta. The coefficients b_1 , b_2 , b_{s^2} , and b_Δ are the Lagrangian bias parameters, which can be thought of as weights corresponding to each of the operators of the density field and are specific to the tracer sample being modeled. Note that the full Lagrangian weighting function used to advect the particles to Eulerian space is $1 + \delta_g(\mathbf{q})$, such that unbiased tracers recover the matter density field.

We construct our tracer density fields based on this model as follows. We start from a random realization of the initial Gaussian field on a $N_{\text{grid}} = 512^3$ grid in a $L = (1 h^{-1}\text{Gpc})^3$ box, matching the reso-

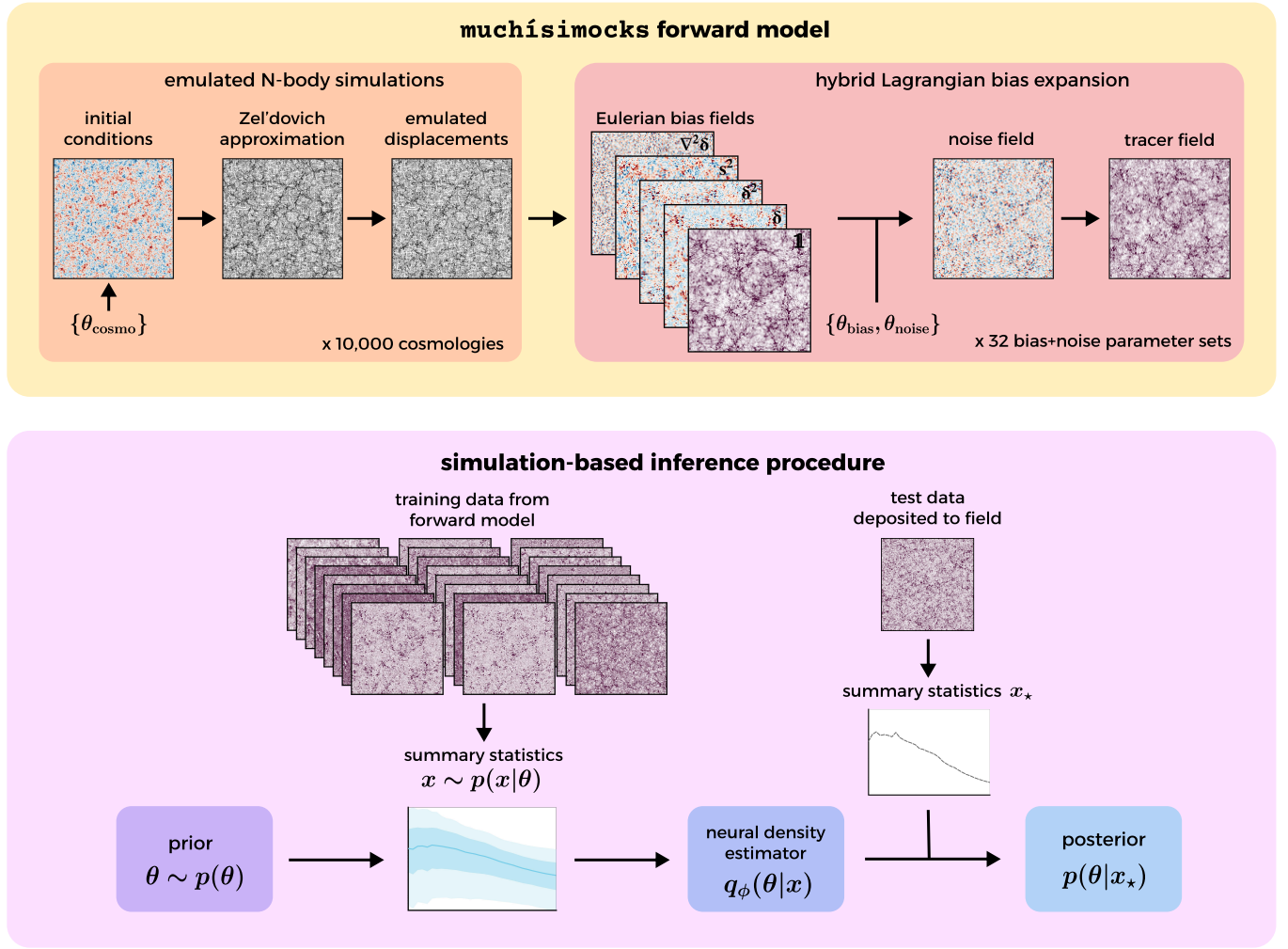


Figure 1. Schematic overview of the muchisimocks suite and simulation-based inference procedure. For our forward model, we start with an initial conditions field based on input cosmological parameters, and apply the Zel'dovich approximation to obtain first-order displacements; these are then passed to the map2map emulator which predicts the N -body displacement field. Using these as inputs to the hybrid Lagrangian bias expansion, we produce second-order Lagrangian bias fields advected to Eulerian space; these compose the core of the muchisimocks suite. These are then combined with a choice of bias parameters, as well as a full-field noise model. The combined tracer fields and associated parameters are used to train a neural posterior estimator, which conditioned on test data can then predict the associated parameters.

lution and size of the Quijote simulations that map2map was trained on, and use the ZA to compute approximate displacements. We pass the ZA displacement and velocity fields as input to map2map (§2.1), which outputs emulated N -body displacement and velocity fields. We compute the Lagrangian operators based on the linear density field, smoothing these with a damping scale of $k = 0.75h\text{Mpc}^{-1}$. We then perform the advection to Eulerian space: The tracers are deposited onto a grid in Eulerian space with a cloud-in-cell (CIC) assignment, with positions given by the simulation displacement field and weights given by Lagrangian operator. This gives five bias fields in Eulerian space, $\{O\}$: weighting the Lagrangian density field by the constant $\mathbb{1}$ gives the homogeneous field; by $\delta_L(\mathbf{q})$, the linear density field; by $\delta_L^2(\mathbf{q}) - \langle \delta^2 \rangle$, the quadratic density field; by $s^2(\mathbf{q}) - \langle s^2 \rangle$, the tidal field; and by $\Delta^2\delta(\mathbf{q})$, the Laplacian. These fields can be weighted by the bias coefficients and summed to produce an overall tracer field δ_g .

We perform two postprocessing steps on these fields. To reduce storage space, we reduce the size of these fields by cutting the high- k

modes: we remove modes beyond $k_{\text{Nyquist}} = \pi N_{\text{grid,cut}}/L \approx 0.4$, where $N_{\text{grid,cut}} = 128$, our reduced grid size. We do this by Fourier transforming each bias field, nulling the modes higher than k_{Nyquist} , and inverse Fourier transforming back to real space. At scales higher than this our hybrid bias model loses accuracy anyway and we don't consider them in our inference, so we do not give up accuracy with this k -cut. We perform the same reduction on the velocity fields, cutting each component of the velocity vectors in k -space. We finally perform a deconvolution of the bias fields, using a linear interpolation kernel at the original grid size N_{grid} to correct for the smoothing by the CIC kernel. (Technically the deconvolution should be performed before the k -cut, but we confirm the results are effectively equivalent and this ordering is much faster.) This deconvolution could be done at the summary statistics level, but we choose to deconvolve at the field level for ease of use.

The final main suite, designed as a training set for cosmological inference, contains 10,000 sets of tracer fields (5 Lagrangian bias fields advected to Eulerian space) and velocity fields, each at a dif-

ferent cosmology; each has volume $(1h^{-1}\text{Gpc})^3$ and grid size 128^3 . We also generate multiple smaller suites for testing, detailed in §3.5. We call these suites *muchisimocks* (a portmanteau of the words “muchísimos”, Spanish for “many”, and “mocks”).

2.3 Modeling noise at the field level

ksf: technically this is a postprocessing step only needed for the SBI part, is it fine here?

We require a model for stochastic noise in order to model realistic discrete galaxy populations. This can be modeled at the level of each summary statistic, but our goal is to make this suite usable for a wide range of statistics as well as field-level analyses; we thus desire a field-level noise model. Further, as we are considering higher-order statistics, we need to include non-Gaussian noise contributions. We thus turn to the noise model of Rubira & Schmidt (2025), an EFT approach described by nonlinear couplings of a single Gaussian field to the bias fields. In this model the galaxy density field can be written as

$$\delta_g(\mathbf{x}) = \sum_{m=0}^{\infty} \sum_{\mathbb{1}, O} b_O^{\{m\}} [\epsilon_G(\mathbf{x})]^m O(\mathbf{x}) \quad (2)$$

where m is the order of the expansion in the noise field, $\epsilon_G(\mathbf{x}) = \mathcal{N}(O, 1)$ is a unit Gaussian field, O are operators of the galaxy density field, and $\{b_{\mathbb{1}}^{\{m\}}, b_O^{\{m\}}\}$ are free coefficients. In our case we limit to second order in the bias expansion, as described in §2.2. Limiting to zeroth order in m reduces to the noiseless case, as in Equation 1. For our SBI analysis, we include terms through first order in the noise expansion, $m = 1$. That gives us bias coefficients $\{b_O^{\{0\}}\} = b_1, b_2, b_{s^2}, b_{\Delta}$, as in Equation 1 (we fix $b_{\mathbb{1}}^{\{0\}} = 1$), and noise coefficients $\{b_O^{\{1\}}\} = A_{\text{homog}}^{\text{noise}}, A_{b_1}^{\text{noise}}, A_{b_2}^{\text{noise}}, A_{b_{s^2}}^{\text{noise}}, A_{b_{\Delta}}^{\text{noise}}$.

We note that limiting the noise model to $m = 1$ means that we exclude noise contributions to $P_{gm}(k)$, as at $m = 1$ the noise field ϵ_G has zero mean (whereas $P_{gg}(k)$ and $B_{ggg}(k_1, k_2, k_3)$ get factors of ϵ_G^2 which don't vanish). This means that the bias parameters may have to absorb partial noise contributions, biasing the inference of the bias parameters themselves; future work could extend this noise model to $m = 2$ or choose a well-motivated subset of noise parameters tailored to the statistics analyzed.

Operationally, we generate one unit Gaussian noise field ϵ_G per cosmology in our suite. We store these and use them to construct the noisy tracer field on-the-fly just before computing the summary statistics of the field. This means that we use the same noise realization when we select different bias and noise parameter sets for the same cosmology (as detailed in §3.2 and §3.5), which introduces some undesirable non-independence; however, this allows for easy reproducibility, and we expect the effect to be minimal, while the storage cost of one noise field per bias model is prohibitive.

3 SIMULATION-BASED INFERENCE METHODOLOGY

Our SBI approach uses neural posterior estimation (§3.1) trained on the main *muchisimocks* Latin hypercube (§3.2) with summary statistics computed on the mocks (§3.3). We determine implementation and training details (§3.4) and apply our final pipeline to a range of test sets (§3.5). An overview of the SBI training and testing procedure is shown in the lower panel of Figure 1.

3.1 Neural posterior estimation

The main idea behind simulation-based inference is that we can infer a posterior distribution of our data over the parameters of interest using only a *simulator*, which produces data samples given a set of parameters. This is distinct from explicit likelihood inference, in which we are required to define a likelihood of our data with respect to the parameters. Neural density estimators (NDEs) present a flexible and robust way to model a target distribution. Here, we use neural posterior estimation (NPE), in which the posterior distribution is directly approximated with an NDE.

To do this, we aim to minimize the distance between the true posterior $p(\theta|x)$ and the NDE model of the posterior $q_{\phi}(\theta|x)$, where ϕ are the weights of the neural network model. We use maximum likelihood estimation, which minimizes the Kullback-Leibler divergence between the distributions. In practice, we consider the posteriors multiplied by $p(x)$, and use the fact that $p(\theta|x)p(x) = p(\theta, x)$:

$$\begin{aligned} \min_{\phi} D_{\text{KL}}(p(\theta|x) \parallel q_{\phi}(\theta|x) p(x)) &= \min_{\phi} \int p(\theta, x) \log \frac{p(\theta|x)}{q_{\phi}(\theta|x)} d\theta dx \\ &\approx \min_{\phi} \sum_i \log p(\theta_i|x_i) - \log q_{\phi}(\theta_i|x_i) \\ &\approx \min_{\phi} \sum_i -\log q_{\phi}(\theta_i|x_i) \\ &\approx \max_{\phi} \sum_i \log q_{\phi}(\theta_i|x_i) \end{aligned} \quad (3)$$

where in the second line, we perform a Monte Carlo expectation of the integral using discrete samples from our simulation (dropping the $\frac{1}{N}$ as it does not affect the minimization); in the third line, we observe that we are minimizing with respect to the neural network weights ϕ and the true posterior $p(\theta|x)$ is independent of ϕ so it drops out; and in the final line we use the fact that minimizing the negative is the same as maximizing the positive.

Normalizing flows are a popular choice for the NDE, thanks to their expressivity and flexibility. The key idea is to transform an initial simple distribution into the target distribution through a chain of transformations. A normalizing flow is defined as

$$q_{\phi}(\theta|x) = \pi_z(f^{-1}(\theta, x)) \left| \det \left(\frac{\partial f^{-1}}{\partial \theta} \right) \right|, \quad (4)$$

where f is an invertible transformation modeled as a neural network and π_z is a base distribution, for which we follow the common choice of a Gaussian, $\pi_z(z) = \mathcal{N}(z|\mathbf{0}, \mathbf{I})$. The Jacobian (the determinant term) arises from the change of variables, and involves f^{-1} , thus requiring f to be invertible. The invertibility property of NFs both enables us to sample from the network and gives us an explicit density, which is key for performing SBI.

Specifically, we use masked autoregressive flows (MAFs), which stack Masked Autoencoders for Distribution Estimation (MADEs) in order to model more complex distributions:

$$q_{\phi}(\theta | x) = \pi_z(z_0) \prod_{t=1}^T \left| \det \left(\frac{\partial f_t^{-1}}{\partial z_t} \right) \right|, \quad (5)$$

where $z_0 \sim \pi_z$ and t indexes the MADE blocks. Each network f_t takes as input the previous variable z_{t-1} as well as the observation x , and the final layer is identified with the parameters of interest, $z_T \equiv \theta$. The loss function derived from the KL divergence (Equation 3) is used to train the network and learn the optimal weights and biases.

Table 1. Uniform prior ranges for the cosmological, bias, and noise parameters. The cosmological parameters are based on a $\sim 10\sigma$ range around the [Planck Collaboration et al. \(2014\)](#) best-fit values, and the bias and noise parameters are loosely based on the hypervolume found by [Zennaro et al. \(2022\)](#) for a wide range of realistic galaxy populations.

Parameter	Lower bound	Upper bound
<i>Cosmological parameters</i>		
Ω_c	0.23	0.40
σ_8	0.65	0.90
h	0.6	0.8
Ω_b	0.04	0.06
n_s	0.92	1.01
<i>Bias parameters</i>		
b_1	-1.0	3.0
b_2	-2.0	2.0
b_{s^2}	-2.0	2.0
b_Δ	-10.0	10.0
<i>Noise parameters</i>		
$A_{\text{homog}}^{\text{noise}}$	-3.0	3.0
$A_{b_1}^{\text{noise}}$	-3.0	3.0
$A_{b_2}^{\text{noise}}$	-2.0	2.0
$A_{b_{s^2}}^{\text{noise}}$	-5.0	5.0
$A_{b_\Delta}^{\text{noise}}$	-10.0	10.0

3.2 Training set

To train our simulation-based inference model, we use the main *muchisimocks* suite of second-order bias fields to construct a set of tracer fields. Our prior on the cosmological parameters is set by the range of the suite, as described in 2.1. We need to define the range of the bias coefficients $\{b\}$ we use to sum the fields; for this, we turn to the results of [Zennaro et al. \(2022\)](#), which define priors that capture a range of realistic galaxy populations modeled with the extended subhalo abundance matching technique ([Contreras et al. 2021](#)). We choose to use uniform priors with slightly wider ranges than the relations found in this work, a more conservative choice; these prior ranges are shown in Table 1. For the noise coefficients $\{A^{\text{noise}}\}$ we choose wide uniform priors loosely based off of the corresponding (noiseless) bias coefficients, with empirical checks ensuring that these vary the statistics within a reasonable range. The prior ranges of the bias and noise parameters used in our SBI analysis are shown in Table 1.

While we are limited to the 10,000 cosmologies defined in the *muchisimocks* suite, we can combine these into tracer fields with any number of bias parameter sets we choose. At a given cosmology, we then have multiple tracer fields generated with distinct sets of bias and noise parameters. Though this means the final tracer fields are not fully independent of each other, this has been shown (e.g. in [Hahn et al. \(2023c\)](#)) to improve inference by significantly augmenting the training data size and allowing the model to disentangle cosmology and bias parameters. Note that the summing of bias fields with the chosen bias coefficients is essentially instantaneous and can be done in postprocessing just before computing summary statistics, and thus requires no additional compute time or storage (though we are still limited by the time required to compute summary statistics on these fields).

As we later want to test convergence of the inference with the number of bias parameter sets, we use a *nested Latin hypercube* to generate the bias parameters. The top-level LH contains 320,000

points, resulting in 32 independent bias parameter sets for each of the 10,000 cosmologies. Nested subsets of this LH are also LHs, with 16, 8, 4, 2, and 1 bias parameter sets per cosmology. We generate a nested LH for the 4 bias parameters and 5 noise parameters, for a total of 9 independently varied parameters.

This was achieved with the [Qian \(2009\)](#) algorithm for nested Latin hypercubes, which works by starting with a two-layer structure, with coarse cells and fine cells on a multidimensional grid. It constructs a standard LH on the coarse cells, and then fills in the finer grid, also ensuring it creates a valid LH. It then performs random permutations of rows and cells that still preserve the LH structure but remove any deterministic structure from the filling procedure, resulting in a random realization of two nested LHs. This is then repeated for subsequent layers by adding finer and finer grids. A python implementation of this algorithm along with validation tests are available in the *gethypercube*¹ package. [ksf: i may include an appendix detailing my nested-LH package! thoughts?](#) Though the LHs at all levels are exact (the samples live in disjoint intervals across all the dimensions), they are not guaranteed to have optimal space-filling properties compared to an optimized single-level LH. Optimizations are possible but very time-consuming, so we use the standard nested hypercube (with the randomization described above); this still achieves LHs that have significantly better space-filling properties across all levels than either taking random subsets of a top-level LH, or stacking bottom-level LHs. The use of LH-distributed parameters has been shown to be important for accurate and robust training, so we expect that this nested strategy leads to robustness in our convergence tests, which are a key result of this work.

To generate our final tracer fields, we weight the *muchisimocks* bias fields and noise field by the selected set of corresponding bias and noise parameters as in Equation (2), and sum these weighted fields. We normalize these fields by the original map2map grid size (512^3) before computing their summary statistics.

3.3 Summary statistics

We perform our SBI analysis with our *muchisimocks* framework on three standard summary statistics: the galaxy–galaxy power spectrum $P_{gg}(k)$, the galaxy–matter power spectrum $P_{gm}(k)$, and the bispectrum $B_{ggg}(k_1, k_2, k_3)$.

The galaxy–galaxy auto power spectrum $P_{gg}(k)$ is defined as

$$\langle \delta_g(\mathbf{k}) \delta_g(\mathbf{k}') \rangle = (2\pi)^3 \delta_D(\mathbf{k} + \mathbf{k}') P_{gg}(k), \quad (6)$$

where $k = |\mathbf{k}|$ is the Fourier space wavenumber. This is the two-point function of the galaxy field, the most common statistic for cosmological inference from galaxy samples, which captures all the Gaussian information. We use 31 logarithmically spaced bins from $k = 0.01$ to $k \approx 0.37$; the maximum k value is set to be below the Nyquist frequency of our mocks. We use the BACCO code to compute $P_{gg}(k)$ for our mocks. We can compute the 15 cross-terms of the 5 bias fields and later sum their contributions based on the bias parameters to obtain the power spectrum of the total tracer field; in practice we do this for the noiseless case, but in the noisy case compute $P_{gg}(k)$ for the total tracer field as this would add additional terms.

The galaxy–matter cross power spectrum $P_{gm}(k)$ is defined by

$$\langle \delta_g(\mathbf{k}) \delta_m(\mathbf{k}') \rangle = (2\pi)^3 \delta_D(\mathbf{k} + \mathbf{k}') P_{gm}(k), \quad (7)$$

where $\delta_m(\mathbf{k})$ is the matter density field in Fourier space. While not

¹ <http://github.com/kstoreyf/gethypercube>

directly observable as it relies on the matter field, this represents the galaxy-galaxy lensing contribution to 3x2pt analyses. We use the same binning as $P_{gg}(k)$, and again use the BACCO code to compute $P_{gm}(k)$ for our mocks; for our fiducial results, we only use up to $k_{\max} = 0.25$ for $P_{gm}(k)$.

The galaxy auto-bispectrum $B_{ggg}(k_1, k_2, k_3)$ is defined by

$$\langle \delta_g(\mathbf{k}_1) \delta_g(\mathbf{k}_2) \delta_g(\mathbf{k}_3) \rangle = (2\pi)^3 \delta_D(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3) B_{ggg}(k_1, k_2, k_3). \quad (8)$$

This is the Fourier space three-point function. We choose 12 linearly spaced bin edges from $k = 0.01$ to $k = 0.4$, yielding 203 triangles; for our fiducial results, we only use triangles with edges up to $k_{\max} = 0.25$, yielding 50 triangles. We use the PolyBin3D code (Philcox & Flöss 2024) to compute $B_{ggg}(k_1, k_2, k_3)$ for our mocks.

The summary statistics computed on the muchisimocks training set are shown in Fig. 2.

3.4 Implementation and training

Before training, we reparameterize the cosmology and bias parameters to combinations that are better constrained by the summary statistics, and thus should result in improved training convergence. We multiply the bias and noise parameters by the relevant power of σ_8 , obtaining $\sigma_8 b_1$, $\sigma_8^2 b_2$, $\sigma_8^2 b_{s^2}$, $\sigma_8 b_\Delta$, $\sigma_8 A_{b_1}^{\text{noise}}$, $\sigma_8^2 A_{b_2}^{\text{noise}}$, $\sigma_8^2 A_{b_{s^2}}^{\text{noise}}$, and $\sigma_8 A_{b_\Delta}^{\text{noise}}$. When sampling from the posterior, we can divide out the appropriate power of the recovered σ_8 to obtain estimates of the original bias and noise parameters. We also do not include the cosmological parameters h , Ω_b , and n_s in the SBI training, as we do not expect to constrain these parameters (and as confirmed in tests); these are still varied in the training set and implicitly marginalized over.

We use the publicly available sbi (Tejero-Cantero et al. 2020) package for the NPE and MAF implementations (Papamakarios & Murray 2016; Greenberg et al. 2019); the MAF uses the nflows Python package (Durkan et al. 2019, 2020). Our data vector is comprised of the binned statistics discussed in §3.3, concatenated when using multiple statics. The statistics combinations we show are: $P_{gg}(k)$, $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3)$, $P_{gg}(k) + P_{gm}(k)$, $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ (we exclude $B_{ggg}(k_1, k_2, k_3)$ and $P_{gm}(k)$ alone, as these do not contain much constraining power on their own and in practice are almost always considered in combination with $P_{gg}(k)$). We choose a 90%/10% training/validation split of our training set. We use the Adam optimizer, and stop training when the mean loss over the validation set does not improve for 20 epochs; we save the network weights at the best validation loss.

We perform a sweep over several hyperparameters, using 30 random samples from the following sets: the learning rate $\{10^{-2}, 3 \times 10^{-3}, 10^{-3}, 3 \times 10^{-4}, 10^{-4}, 3 \times 10^{-5}, 10^{-5}\}$, hidden features $\{32, 64, 128, 256\}$, training batch size $\{32, 64, 128, 256\}$, and number of transforms $\{4, 6, 8, 10\}$. The sweeps are performed separately for each statistic or combination of statistics, and for each choice of noiseless (bias parameters only) and noisy (bias and noise parameters) training data; we use the maximum size training set of 320,000 data vectors, and for the statistics use $k_{\max}^{P_{gg}} = 0.37$, $k_{\max}^{P_{gm}} = 0.25$, and $k_{\max}^B = 0.25$. We look at the ordered mean validation losses of the 30 trained models for each statistics combination, and observe a significant drop for the top three models (this number will vary depending on the number models explored in the sweep and is generally chosen empirically). We thus choose to use an equally weighted ensemble of the top three MAFs when performing inference, which has been shown to lead to more stable results (e.g. Alvey et al. 2026); explicitly, we sample the MAFs as

Table 2. Overview of the muchisimocks mock suites, as well as additional out-of-distribution test mocks. The label mm indicates the muchisimocks forward model; LH is a Latin hypercube; Planck is the Planck Collaboration et al. (2014) cosmology. For the LHs, the prior ranges for the parameters are those in Table 1. N_{bias} indicates the total number of bias and noise parameter sets; the training set uses 32 per cosmology, while the coverage and cosmic variance sets use 1 per cosmology.

Suite	$N_{\text{realizations}}$	$N_{\text{cosmologies}}$	N_{bias}
mm training set	10,000 (LH)	10,000 (LH)	320,000 (LH)
mm coverage test set	1,000	1,000 (LH)	1,000 (LH)
mm cosmic variance test set	1,000	1 (Planck)	1 (matched to SHAMe $\bar{n} = 2.2 \times 10^{-4}$)
BACCO + SHAMe mock	1	1 (Planck)	3 ($\bar{n} = \{1.1, 2.2, 5.4\} \times 10^{-4}$)

$q_\phi(\theta | x) = \frac{1}{3} \sum_{j=1}^3 q_\phi^j(\theta | x)$. We use the sets of hyperparameters from the three top MAFs to train variations of the model for the given statistics combination, such as when varying the number of training points and the $k_{\max}$ values, rather than running dedicated sweeps which would be computationally very expensive. The set of best identified hyperparameters vary across the statistics, with the noisy case (which varies more parameters) generally preferring larger hidden feature vectors and batch sizes, though most learning rates in both noisy and noiseless cases are around 10^{-4} .

Training one MAF on the maximum size training set of 320,000 data vectors takes a few hours on a single CPU for the fiducial model described above. To display the posteriors and compute covariances and other statistics from them, we draw 10,000 samples from the posterior; we confirm that the posteriors are generally converged with this number of samples.

3.5 Test sets

We generate several test sets to validate our muchisimocks suite and our SBI approach. An overview of these test suites, as well as the training set, is shown in Table 2.

Our first test set is a *cosmic variance set*. For this we use the muchisimocks pipeline to construct 1,000 additional mocks each with a different random seeds, all at the same fixed cosmology; we choose this to be the same cosmology as the out-of-distribution SHAMe mock test described below: $\Omega_c = 0.309$, $\sigma_8 = 0.816$, $h = 0.677$, $\Omega_b = 0.0486$, and $n_s = 0.967$. We fix the bias parameters to those measured from that mock, and use these to combine the bias fields into a tracer field for each realization. For the noise parameters, we cannot measure these from the mock directly (they are very poorly constrained); to identify realistic noise parameters, we generate 1,000 sets of noise parameters and add the resulting noise field to each of the 1,000 mocks, obtaining tracer fields that have the same cosmology and bias parameters but different noise parameters. We compute the statistics of these fields, and then compute the reduced χ^2 between them and the statistics of the SHAMe mock with fiducial number density $\bar{n} = 2.2 \times 10^{-4} (h^{-1} \text{Mpc})^{-3}$. Specifically, we use $P_{gg}(k)$ and $B_{ggg}(k_1, k_2, k_3)$ with the fiducial $k_{\max}$ choices (the noise model does not affect $P_{gm}(k)$, as discussed in §2.3, so we exclude it), and choose the set of noise parameters that gives the lowest χ^2 . We then use this single set of SHAMe-mock-matched noise parameters to compute the noise terms of our model, using 1,000 different Gaussian noise field realizations ϵ_G , and combine this with the noiseless term from only the bias fields to obtain our final set of 1,000 cosmic variance

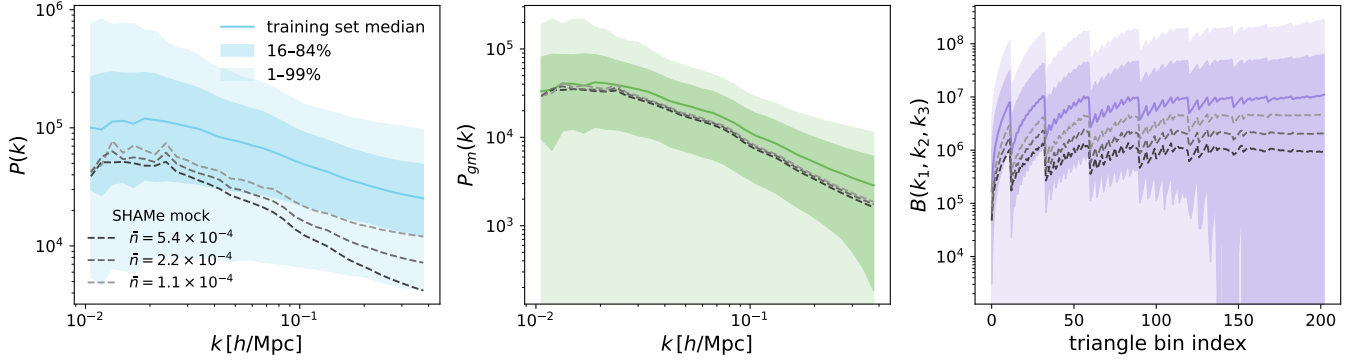


Figure 2. The summary statistics computed on the *muchisimocks* training set: the galaxy–galaxy power spectrum $P_{gg}(k)$, the galaxy–matter power spectrum $P_{gm}(k)$, and the galaxy bispectrum $B_{ggg}(k_1, k_2, k_3)$. The median, 16th–84th percentiles, and 1st–99th percentiles of the training set are shown. Out-of-distribution test mock catalogs constructed with extended subhalo abundance matching (SHAME) on a BACCO simulation is also shown, for three number densities (dashed). In our fiducial inference model, scales below $k_{\max} = 0.25$ for $P_{gm}(k)$ and triangles with sides smaller than $k_{\max} = 0.25$ for $B_{ggg}(k_1, k_2, k_3)$ are excluded.

mocks at fixed cosmology, bias, and noise parameters, only varying the random seeds.

Our second test set is our *coverage set*. This consists of 1,000 mocks generated with the *muchisimocks* pipeline that span the same prior range of cosmology, bias, and noise parameters as in the training set. To select the parameters, we construct an independent LH of size 1,000 for the cosmologies, and a separate LH for the bias and noise parameters, with a single set of bias and noise parameters per cosmology for a total of 1,000 mocks. The coverage set tests the performance of our method across the prior space and its dependence on the parameter values.

Our final test set is our *SHAME mock*; we construct a realistic catalog of galaxies using the extended subhalo abundance matching (SHAME) approach (Contreras et al. 2021) applied to a BACCO N -body simulation Angulo et al. (2021). This is a stringent test as it uses both an independent, significantly higher resolution underlying cosmological simulation and a completely different galaxy formation model than the *muchisimocks* mocks used to train the inference network, testing the generalizability of our model. In detail, we use a simulation with a box size of $(1024 h^{-1}\text{Mpc})^3$ (slightly larger than the *muchisimocks* training data) and 3072^3 particles, which uses the fixed-and-paired technique (Angulo et al. 2021), and has Planck cosmology $\Omega_c = 0.309$, $\sigma_8 = 0.816$, $h = 0.677$, $\Omega_b = 0.0486$, and $n_s = 0.967$ (Planck Collaboration et al. 2014). Halos are identified using a variant of the SUBFIND halo finder (Springel et al. 2001), and we populate these halos with galaxies using the SHAME model with the following parameter choices: $\sigma_{\log M} = 0.47513517$, $t_{\text{merger}} = -0.4577029$, $f_{k,\text{cen+sat}} = 0.20147321$, $f_{k,\text{cen-sat}} = -0.02964998$, $\beta_{\text{lum}} = 14.82600407$. *ksf: think it's ok to use fewer sigfigs here?* We choose a fiducial number density of $\bar{n} = 2.2 \times 10^{-4} (h^{-1}\text{Mpc})^{-3}$, similar to that of DESI luminous red galaxies (LRGs) when scaled to $z = 0$. *ksf: checking that this is right? also are the shame parameters chosen to be lrg-like?* To check our inference results, we measure the bias parameters of this sample using the field-level probabilistic bias approach of Stücker et al. (2025b,a); we find $b_1 = 0.474$, $b_2 = 0.127$, and $b_{s,2} = -0.339$ (we do not expect b_{Δ} to be well-measured by this approach). We also test a higher and lower number density sample, of $5.4 \times 10^{-4} (h^{-1}\text{Mpc})^{-3}$ and $1.1 \times 10^{-4} (h^{-1}\text{Mpc})^{-3}$; these are measured to have slightly different bias parameters. The summary statistics computed on the SHAME mock are shown in Fig. 2; they largely fall within the middle 68% of the training set, with the ex-

ception of $P_{gg}(k)$ at smaller scales, which reaches the edge of the training set distribution but still falls within it.

For each of the two simulations of opposite phases from the fixed-and-paired pair, the SHAME model gives a set of discrete galaxy tracers. We deposit these using the Cloud-in-Cell mass assignment scheme to a mesh of size 524^3 to achieve approximately the same cell size as the *muchisimocks* mocks given the slight difference in box size. We then remove the high- k modes to downsample to 132^3 , again to match the *muchisimocks* cell size, and deconvolve the field in the same way as the *muchisimocks* fields, based on the original grid size. After turning these into normalized overdensity fields, we compute the statistics on each of the two fields with the same method used for the training data, and take the mean of these two statistics as the data vector for inference.

4 SBI RESULTS

4.1 In-distribution recovery tests

4.1.1 Cosmic variance test set results

We first validate our SBI pipeline on the cosmic variance test set, at fixed values of the cosmology, bias, and noise parameters. We compute summary statistics on the test data in the same way as on the training data.

For the cosmic variance test set, we compute the mean of the statistics computed on the 1,000 individual realizations, and use this as the data vector for inference. In Appendix A, we show that this gives consistent posteriors as combining the results of inference on each of the 1,000 mocks, with the mean and MAP closely agreeing; the posterior widths do differ by up to 5%, but we expect the conclusions from this validation test to hold. This inference on the mean has the benefit of averaging out the effects of cosmic variance to obtain the mean constraint and allowing us to visualize a typical contour, as well as being significantly faster. The posteriors are shown in Figure 3 for three key parameters, Ω_m , σ_8 , and b_1 ; we infer and marginalize over the other bias parameters and the noise parameters. (The full contour plot is shown in Fig. B1.)

While $P_{gg}(k)$ on its own provides significant constraining power on Ω_m , it results in a weak constraint on b_1 and a weak, biased constraint on σ_8 . The weakness of these constraints is expected as

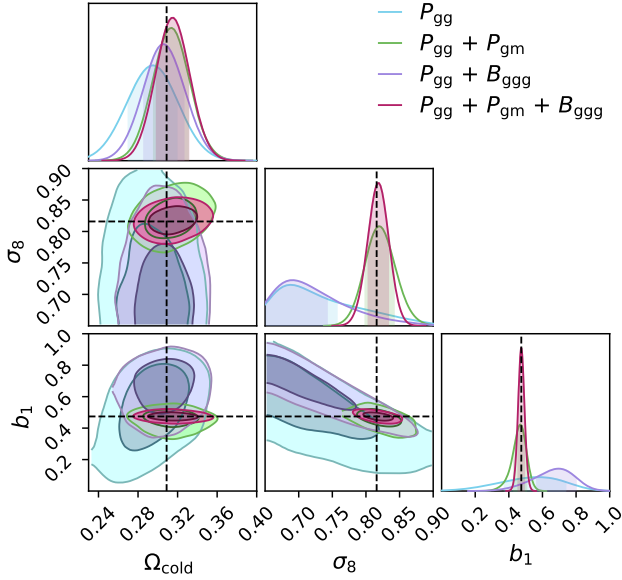


Figure 3. Posteriors for the key parameters Ω_m , σ_8 , and b_1 inferred from the mean of the statistics over our fixed-parameter cosmic variance test set, for each of the statistics combinations considered. The inclusion of $P_{gm}(k)$ breaks the $\sigma_8 - b_1$ degeneracy, providing tight and unbiased constraints on the parameters; adding $B_{ggg}(k_1, k_2, k_3)$ tightens the constraints. The other bias parameters and the noise parameters are inferred and marginalized over; the full posterior plot is shown in Fig. B1.

a result of the perfect degeneracy between σ_8 and b_1 in the power spectrum (as $P_{gg}(k) = (b_1 \sigma_8)^2 \hat{P}(k)$); we check that the combination $\sigma_8 b_1$ is accurately constrained. Including $B_{ggg}(k_1, k_2, k_3)$ interestingly only slightly increases the precision on Ω_m ; further, it does not improve the bias on σ_8 or b_1 . Notably, it does significantly increase the precision on the higher-order bias parameters b_2 and b_{s^2} . We see that using the statistics combination $P_{gg}(k) + P_{gm}(k)$ produces unbiased constraints on σ_8 and b_1 ; this is expected as $P_{gm}(k)$ is able to break their degeneracy (as $P_{gm}(k) = b_1 \sigma_8^2 \hat{P}(k)$). This corroborates our hypothesis that the bias observed in σ_8 for $P_{gg}(k)$ and $B_{ggg}(k_1, k_2, k_3)$ is likely a result of projection effects caused by a lack of constraining power. The full statistics combination $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ preserves the unbiased constraints, and the bispectrum adds an additional improvement of constraining power compared to $P_{gg}(k) + P_{gm}(k)$: it improves Ω_m by 6%, σ_8 by 49%, and b_1 by 2.1×; it has an even stronger effect on the bias parameters, improving b_2 by 3.5×, b_{s^2} by 2.9×, and b_Δ by 2.5×. The noise parameters are all poorly constrained, recovering the prior; however, we check that marginalizing over these is necessary and sufficient for unbiased constraints.

These tight, unbiased constraints on the cosmology and bias parameters with the full data vector considered serve as a validation of our inference approach. We note that these results are for real space, and we would expect differences in the relative information content in the statistics in redshift-space, where there is additional cosmological information contained in redshift-space distortions.

4.1.2 Coverage test set results

We infer the parameters of the 1,000 samples of our coverage test set, spanning the prior range of our training set; this allows us to test

the dependence of the estimated parameters on the true parameter values. Note that the prior has a wide range of values of the noise parameters, so these test samples will have a variety of noise levels.

In Figure 4, we show the differences between the predicted values and true values of the three key parameters parameters as a function of the true value; we display the median of the 1,000 mocks, and the 16-84th percentile range. The predicted value is the mean of the inferred posterior; we check that the behavior is very similar for the maximum a posteriori (MAP) values. For $P_{gg}(k)$, we see that for all of the parameters there is a bias that increases with distance from the center of the prior, with a bias high at smaller true values and low at higher values. This is expected when the statistic does not provide significant constraining power on the parameter, as is the case with $P_{gg}(k)$, as the prediction will tend towards the mean. Including $B_{ggg}(k_1, k_2, k_3)$ improves the parameter-dependent bias for all parameters, and also generally tightens the range of the residuals. For $P_{gg}(k) + P_{gm}(k)$, we can see that the bias on σ_8 improves significantly, as $P_{gm}(k)$ breaks the degeneracy with b_1 and thus provides stronger constraining power on all parameters and avoids projection effects, as we saw in the cosmic variance test set. $P_{gm}(k)$ also improves the accuracy and precision of the constraints on Ω_m and b_1 . Further including $B_{ggg}(k_1, k_2, k_3)$ helps marginally for all parameters, and this full parameter combination $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ yields largely unbiased results across the coverage set, indicating that our inference is robust across the prior. There are still slight biases for $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ at the edges of the prior range, particularly at high Ω_m and σ_8 , so the inference should be considered to be most robust with a small buffer from the prior edges.

The residuals shown above only quantify the accuracy of the means of the posteriors; to assess the inferred uncertainties, we compute the coverage of the samples, defined as the fraction of samples for which the true value falls within the predicted uncertainty range, as a function of the given level of uncertainty. For instance, for a credibility level of 68%, 68% of the samples should have the truth fall within their predicted 68% error region (“empirical coverage”). A higher empirical coverage compared to credibility level indicates underconfidence, while lower indicates overconfidence. This coverage (or probability-probability) plot is shown in Figure 5, for the 500 mocks of the coverage set closest to the center of the prior in Ω_m , σ_8 , and Ω_m . We see that the inferred uncertainties are generally quite well calibrated for all three key parameters. Using only $P_{gg}(k)$ or $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3)$ results in a slight underconfidence in the posteriors, at the level of 0.1 – 0.25 σ (assuming a Gaussian distribution centered on the truth). The full statistics combination $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ results in improved calibration of the uncertainties, with only slight fluctuations. We note that for this check, we have excluded the mocks on the edges of the prior to isolate the assessment of the SBI-inferred uncertainties; when we make the coverage plot for all 1,000 mocks of the coverage set, we do see some overconfidence. This is a result of the biases towards the edges of the prior which cause the uncertainty regions to not sufficiently cover the truth, and the uncertainty estimates do not fully compensate for this bias.

These results show that our inference pipeline can accurately recover the cosmological and bias parameters across a range of parameter values.

4.2 Out-of-distribution recovery test: SHAMe mock

The results of our SBI pipeline on our SHAMe out-of-distribution test mock with fiducial number density $\bar{n} = 2.2 \times 10^{-4} (h^{-1} \text{Mpc})^{-3}$

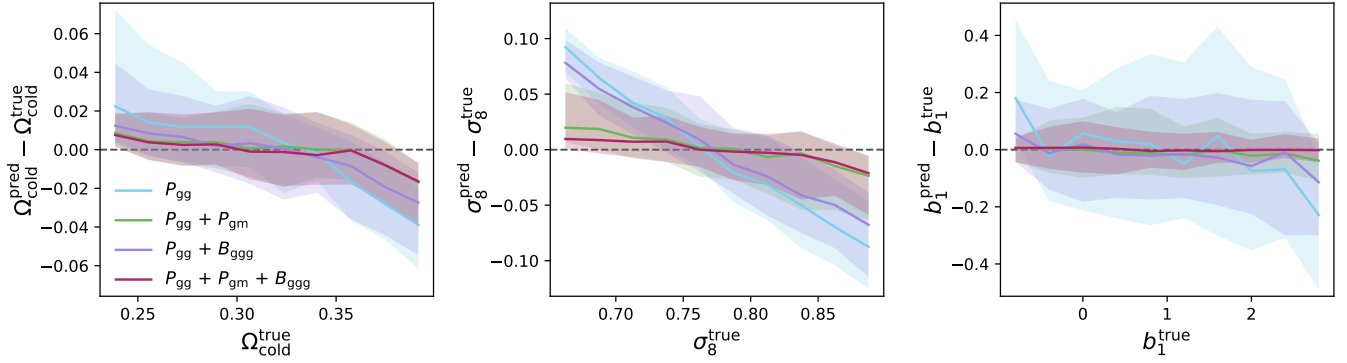


Figure 4. The difference between the predicted (mean of the posterior) and true parameter values for the key parameters for each of the 1,000 coverage test set mocks, as a function of the true value. The median and 16th–84th percentile ranges in bins of the true value are shown for each of the statistics combinations considered. Using the full statistics combination gives largely unbiased recovery across the prior range, with small biases at the high ends of Ω_m and σ_8 ; using fewer statistics results in stronger parameter-dependent biases that tend towards the mean. Including more statistics also results in tighter parameter constraints, most notably with $P_{gm}(k)$ significantly improving the precision and bias on all key parameters.

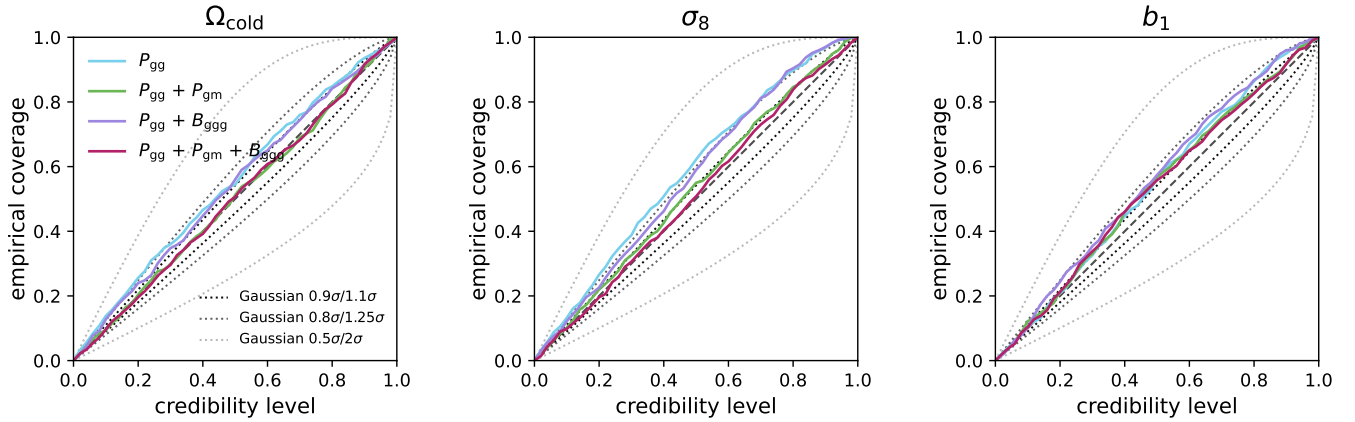


Figure 5. Coverage (probability–probability) plots for the key parameters for the 500 mocks of the coverage test set closest to the center of the prior. The empirical coverage is the fraction of test set points with predicted uncertainties at the given credibility level that encompass the true value. The closeness to the 1:1 line indicates well-calibrated uncertainties; dotted lines show the expected deviations if posterior widths were off by the given factor of σ , assuming Gaussians centered on the truth. The posteriors are largely well-calibrated, with the full statistics combination $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ giving the best calibration across parameters.

are shown in Figure 6 for the key parameters (the full contours are shown in Figure B2). We obtain generally unbiased constraints on the cosmological and bias parameters when using the full statistics combination, with Ω_m , σ_8 , and b_1 all within 1σ from the true values. The trends for the statistics generally follow those seen in the cosmic variance test set (§4.1 and Figure 3), with $P_{gm}(k)$ breaking the degeneracy between σ_8 and b_1 , and $B_{ggg}(k_1, k_2, k_3)$ improving constraints significantly for the bias parameters and moderately for the cosmological parameters. For the SHAME data, we obtain a 34% improvement in precision on Ω_m , a 4× improvement on σ_8 , and a 4.5× improvement on b_1 when including $P_{gm}(k)$ compared to $P_{gg}(k)$ alone; additionally including $B_{ggg}(k_1, k_2, k_3)$ results in 12%, 11%, and 70% improvements on Ω_m , σ_8 , and b_1 respectively compared to $P_{gg}(k) + P_{gm}(k)$.

We do see moderate biases in the values of b_2 and b_{s2} when using the full statistics combination, at the 2.6σ and 1.9σ levels, while these were on average accurately constrained in the cosmic variance test set. This is likely a result of the bias parameters absorbing some

model mismatch, which could be an issue for approaches such that rely on accurate bias parameters such as imposing additional priors on them; however, the cosmological parameters remain unbiased.

We show the results for the full statistics combination $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ on the SHAME samples with a range of number densities in Figure 7. We find that the constraints on the cosmological parameters are robust to the noise level across this range. The constraints on the bias parameter b_1 do lose precision at lower number densities; this could be related to degeneracies with the noise parameters. We note that the noise parameters themselves are poorly constrained in all cases, as seen for the fiducial number density in the full contour plot (Figure B2). However, we check that the noise model is necessary for properly inferring the parameters; assuming noiseless training data results in strong biases in the inference.

We also show the results for inference on the mean data vectors of the cosmic variance sample, which was generated with bias and noise parameters matched to the $\bar{n} = 2.2 \times 10^{-4} (h^{-1} \text{Mpc})^{-3}$ SHAME sample. The precision of the predicted parameters is similar to that

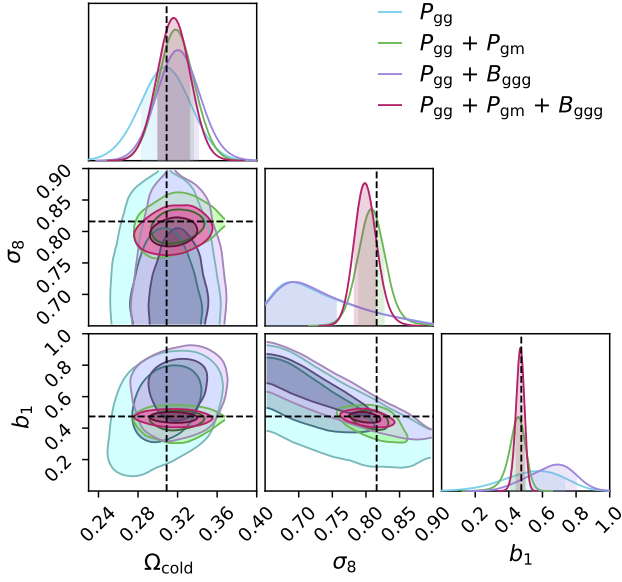


Figure 6. Posteriors for the key parameters Ω_m , σ_8 , and b_1 for our out-of-distribution SHAME mock with fiducial number density $\bar{n} = 2.2 \times 10^{-4} (h^{-1}\text{Mpc})^{-3}$, for each of the statistics combinations considered. The recovered parameters with the full statistics combination are unbiased within 1σ , and the statistics show similar constraining power as in the in-distribution test. The full posterior plot is shown in Fig. B2.

of the actual SHAME sample, with somewhat tighter constraints for the in-distribution mock. This demonstrates that our cosmic variance sample matched to the SHAME mock is a good representation of the information content of a sample with those properties. As it is generated with the same forward model as the training set, comparing it to the SHAME mock results allows us to isolate the effect of model misspecification. Indeed, the bias in σ_8 for the SHAME mock disappears, indicating that the cause is model misspecification rather than remaining projection effects.

We additionally perform a test in which we generate a cosmic variance test set with the SHAME bias parameters for the fiducial number density sample, but no noise. We construct a noiseless training set, with the same bias fields but an independent nested LH that varies only the 4 bias parameters, and train a new SBI model. When applying this model to the noiseless test set, we find that the constraints are consistent with the noisy case for Ω_m , but improve significantly for σ_8 and b_1 . This demonstrates the constraining power lost to noise for the number density considered, even with perfect knowledge of the noise model.

These tests demonstrate the generalizability of our forward model to realistic galaxy populations constructed with completely different underlying models.

4.3 Scale dependence

We investigate the dependence of our results on the maximum k scale included in our inference, for each of the statistics. We keep the maximum scale for $P_{gg}(k)$ fixed to $k_{\text{max}} = 0.37 h\text{Mpc}^{-1}$, a bit below the Nyquist frequency of the box. We expect the galaxy-galaxy power spectrum to be accurate down to this scale: there the map2map emulator of N -body displacements is shown to achieve significantly sub-percent accuracy ($\sim 0.2\%$) (Jamieson et al. 2023),

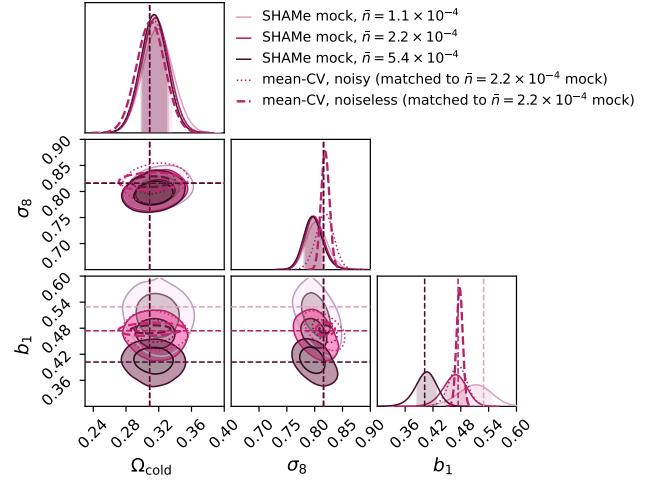


Figure 7. Posteriors for $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ for the key parameters, for varying number density and noise models. The filled contours show the results for the SHAME OOD mock data with different number densities. The dotted contours show inference on the mean of the cosmic variance test set, which was matched to the fiducial $\bar{n} = 2.2 \times 10^{-4} (h^{-1}\text{Mpc})^{-3}$ SHAME mock; the dashed contour shows that same set but generated without any noise, and inferred assuming noiseless data.

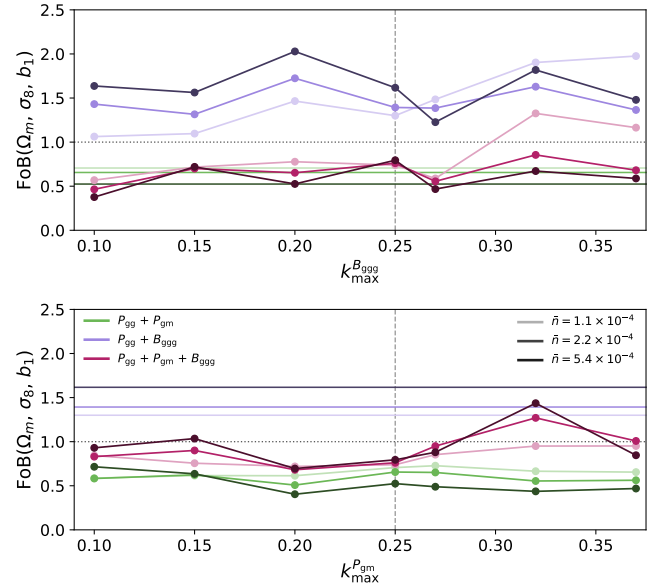


Figure 8. Scale dependence of the figure of bias (FoB) for the three key parameters combined for the SHAME out-of-distribution mock, with varying number density. The top panel shows the FoB as a function of the k_{max} used for the bispectrum, fixing to our fiducial $k_{\text{max}} = 0.25$ for $P_{gm}(k)$; the bottom panel shows the FoB as a function of the k_{max} used for $P_{gm}(k)$, fixing to our fiducial $k_{\text{max}} = 0.25$ for the bispectrum. In both cases, the FoB for the full statistics combination remains around or below 1σ until $k_{\text{max}} = 0.25$, after which the inference shows a bias; we thus use $k_{\text{max}}^{P_{gm}} = k_{\text{max}}^{B_{ggg}} = 0.25$ for our fiducial inference.

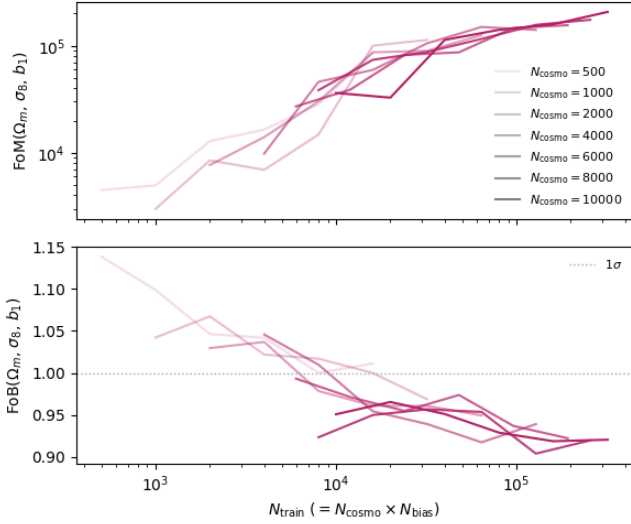


Figure 10. Convergence of the inference with training set size, for $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ on the coverage test set for the three key parameters Ω_m , σ_8 , and b_1 combined. The top panel shows the figure of merit (FoM), and the bottom panel shows the figure of bias (FoB), normalized by the 68th percentile of the χ^2 distribution for three dimensions (corresponding to the three parameters); the median values over the coverage set are shown. The opacity indicates the number of training cosmologies. Increasing the number of training points increases the precision significantly until $N_{\text{train}} \sim 20,000$ and then more marginally, with the number of cosmologies and bias parameter sets contributing similar precision; the bias also plateaus around $N_{\text{train}} \sim 20,000$, with the number of cosmologies mattering more than the number of bias parameters for smaller N_{train} . *ksf: make legends, axis labels bigger ksf: note: this plot is with an older base model, with kmax=0.4 and no ensembling; running the new version right now*

as

$$\text{FoM} = \frac{1}{\sqrt{\det C}}, \quad (10)$$

which is the inverse of the volume of the allowed region, meaning that a higher FoM indicates tighter constraints. Again we compute C only across the subspace considered; in one dimension, this reduces to $\text{FoM}_{1D} = 1/\sigma_\theta$.

Figure 10 shows the figure of merit and figure of bias for the three key parameters combined, averaged over the coverage set, for the full statistics combination as a function of the total number of training points $N_{\text{cosmo}} \times N_{\text{bias}}$, shown as separate lines for each choice of N_{cosmo} . We see that the FoM increases with increasing number of training points, steeply until $N_{\text{train}} = 10,000$ and then more moderately. The FoM does not plateau even at 320,000 training points, indicating that the inference is still not converged and the precision would continue to improve with additional training data. The FoB similarly reduces sharply until $N_{\text{train}} = 10,000$ where it crosses $\text{FoB} = 1\sigma$, but then more-or-less plateaus, indicating that above a few tens of thousands of training points, the training data is sufficient for unbiased inference.

We also find that there is not a strong trend with the number of training cosmologies, as shown by the lines for the different cosmologies largely overlapping. With a smaller number of training data points (below $N_{\text{train}} = 20,000$), it is more important to have more cosmologies rather than more bias parameter sets, shown by the spread in the FoB around that value; however, beyond $N_{\text{cosmo}} = 2000$, it is sufficient to add more bias parameter sets per cosmology. This some-

what surprising finding is encouraging given that generating training data at varying cosmologies is more expensive than populating each cosmology with varying bias parameter sets. Previous SBI analyses already use around this number of training cosmologies with full N -body codes (without the need for emulators, e.g. the Quijote Latin hypercube); our result shows that this may be sufficient in terms of cosmologies, but that additional constraining power and robustness could be gained by populating these with many more galaxy formation parameter sets per cosmology.

In Figure 11, we show the FoM and FoB for all the statistics combinations, for each key parameter individually. Note that for a Gaussian sample, we would expect $\text{FoB} \approx 0.8$; our FoB stays slightly higher than this for all the parameters, bottoming out at 0.85–1.0, possibly as a result of non-Gaussian behavior. We see that for σ_8 , a small number of training cosmologies already saturates the precision for $P_{gg}(k)$ and $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3)$, though around $N_{\text{train}} = 20,000$ are needed to reach a plateau in the FoB below 1σ . Similarly for b_1 for those statistics combinations, the precision begins to plateau around the same number of training points, and seems to saturate around $N_{\text{train}} = 100,000$. The statistics combinations $P_{gg}(k) + P_{gm}(k)$ and $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$ show steep improvements in both precision and accuracy until $N_{\text{train}} = 10,000$ for all parameters, and then improve more moderately. Notably, by our maximum number of training points $N_{\text{train}} = 320,000$, the FoM has not yet plateaued, indicating that additional training data may continue to improve the inference and we have not yet extracted the full information content the statistics. We also again find that for all parameters, increasing the number of bias parameter sets per cosmology results in similar FoM and FoB values as increasing the number of cosmologies, beyond a few thousand cosmologies.

5 DISCUSSION

The results of this analysis outline the successes and challenges of SBI with the hybrid Lagrangian bias expansion and emulated N -body displacements. While we showed that our model is able to provide robust inference on cosmology, model extensions and adjacent analyses will add to our understanding of the information content in galaxy clustering and optimal approaches to extract it.

We have used Lagrangian bias expansion to second order; it is possible that extending into the third order may be necessary for more accurate modeling in the small-scale regime *ksf: citation for this / more to say?*. For the stochastic noise, we have used a noise model based on a Gaussian field coupled to the bias fields following Rubira & Schmidt (2025), to first order in the noise field. This is a more sophisticated model than a standard Gaussian field, or parametric noise models specific to each statistic; however, this still does not capture the noise in $P_{gm}(k)$ and may not be sufficient for other high-order statistics. Extending this noise model to higher order (at the expense of adding additional parameters) or testing other noise models may be important for ensuring the most robust inference.

While the map2map emulator is shown to be accurate for the galaxy–galaxy power spectrum down to small- k , there are remaining issues for using it for cosmological inference. Pellejero Ibanez et al. (2023a) found that when employing it for the hybrid bias expansion, there were biases at intermediate scales; while these were absorbed by noise nuisance parameters, inference could be sensitive to this depending on other modeling choices. The emulator also has not been robustly validated for the whole range of higher order statistics or the full tracer field. Improving the displacement emulator or further validating it will be key for future analyses.

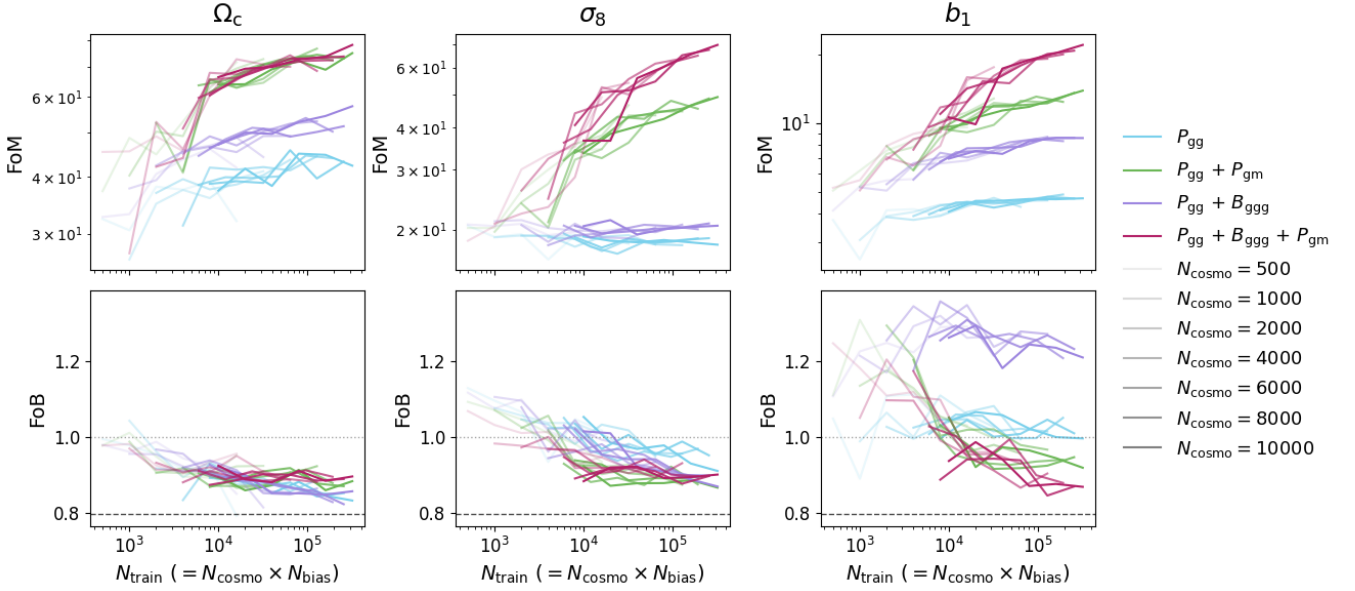


Figure 11. Convergence of the inference with training set size, for all the statistics combinations on the coverage test set for the three key parameters Ω_m , σ_8 , and b_1 individually. The top row shows the figure of merit (FoM) and the bottom row shows the figure of bias (FoB); the median values over the coverage set are shown. The opacity indicates the number of training cosmologies, and the dashed line indicates the expectation for an unbiased Gaussian distribution. The FoM increases with training set size for most statistics and parameters, with the exception of σ_8 with $P_{gg}(k)$ and $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3)$ which do not show improvement; all have an FoB around or below 1σ . *ksf: note: this plot is with an older base model, with kmax=0.4 and no ensembling; running the new version right now*

Bairagi et al. (2025) explored the question of the required number of simulations for cosmological inference, using a large suite of 32,768 N -body simulations called the Big Sobol Sequence. Using neural posterior estimation on the nonlinear dark matter power spectrum down to $k_{\max} = 0.8$, they found that around 4,000 simulations were required to saturate the information content, significantly more than the 2,000-simulation Quijote suite. This stands somewhat in contrast to our finding, that on the one hand 2,000 cosmologies provide comparable results to 10,000 given an equal total number of simulations (including multiple galaxy bias parameter sets), and on the other even 320,000 total training data points does not seem to saturate in precision. However, our convergence findings are not directly comparable, as their work analyzes a matter field statistic and does not sample galaxy bias parameters, so we would expect that fewer simulations are needed. Tucci & Schmidt (2024) also analyze the convergence of their EFT-based SBI approach with the number of simulations, finding that at fixed cosmology 10^5 simulations are required to converge in the inference of the bias parameters. In the free-cosmology case with one cosmological parameter, the inference is slower to converge, and is not clearly fully converged at their maximum 5×10^5 simulations; this is in line with our results that upwards of 10^5 simulations are required for convergence. Further work with large simulation suites in various regimes and different modeling choices is necessary to provide a complete picture of the convergence with the number of simulations.

The muchisimocks approach has the potential to be expanded to answer further questions about the information content in galaxy clustering and the requirements for SBI. With the N -body emulator costing only minutes and the summation of the bias fields with a choice of bias parameters being nearly instantaneous, even more mocks could be generated to understand where the information in various statistics saturates, as we found that some continue to rise

with our full 320,000-mock suite. We could also straightforwardly generate larger volumes with this approach, as current survey volumes are easily outpacing our ability to model these volumes with traditional simulations. With modifications to the forward model, beyond- Λ CDM cosmologies could be explored to understand the sensitivity of various statistics to alternative cosmological models.

A key potential extension is to explore other statistics of the tracer fields to understand their information content. Optimal summary statistics that capture the information content of the full field could be learned, to explore the maximal information in cosmological fields. As we also forward model the velocity fields, this suite could be useful for exploring velocity reconstruction approaches. One could also investigate the effect of more restrictive priors on the Lagrangian bias parameters, encoding our knowledge specific galaxy samples.

The SBI approach with the muchisimocks model that we have demonstrated could be applied to more realistic data, and eventually to real data to perform a cosmological analysis. This analysis has been entirely in real-space, and a redshift-space analysis is forthcoming. Survey realism, including systematics, could be applied to the fields to understand the information content in more realistic settings. Finally, an analysis on survey data such as BOSS or DESI has the potential to provide competitive and complementary results.

6 SUMMARY AND CONCLUSIONS

In this work, we have presented a forward model for galaxy tracer fields that balances robustness and speed, and is useful for simulation-based inference approaches to cosmological inference. The model uses the hybrid Lagrangian bias expansion to second order, based on emulated particle and velocity displacement fields from the map2map emulator of Jamieson et al. (2023) trained on the Quijote mocks,

to generate sets of bias fields that can be weighted and summed to produce tracer fields. We also include a full-field noise model of Rubira & Schmidt (2025) based on a Gaussian field coupled to the bias fields, which we find to be critical for unbiased inference. With this forward model, a $(1h^{-1}\text{Gpc})^3$ mock can be generated at any cosmology in around 8 minutes. We publicly release a suite of these bias fields, called *muchisimocks*, at $z = 0$ in both real- and redshift-space. Our main mock set consists of a Latin hypercube of 10,000 different cosmologies; we also release an independent 1,000-cosmology Latin hypercube “coverage” test set, and a 1,000-realizations “cosmic variance” set at fixed cosmology with varying seeds.

We perform simulation-based inference using use neural posterior estimation. We generate training data from our forward model, recombining each bias field set at a given cosmology with up to 32 bias and noise parameter sets, for a total of 320,000 training mocks. The summary statistics considered are the galaxy–galaxy power spectrum $P_{gg}(k)$, the galaxy–matter cross-spectrum $P_{gm}(k)$, and the galaxy bispectrum $B_{ggg}(k_1, k_2, k_3)$. In addition to the in-distribution tests sets described above, we apply our model to a realistic out-of-distribution mock built with the extended Subhalo Abundance Matching (SHAMe) model on a high-resolution BACCO simulation. With this approach, our findings are the following:

- Inference with $P_{gg}(k)$ alone suffers from strong projection effects, resulting in biased σ_8 even for the in-distribution test sets; this holds even when additionally including $B_{ggg}(k_1, k_2, k_3)$. Including $P_{gm}(k)$, which is related to the galaxy–galaxy lensing observable, results in unbiased inference of Ω_m , σ_8 , and b_1 , as it breaks parameter degeneracies and ameliorates projection effects (see Figs. 3 and 4).
- We recover unbiased cosmological parameters on the out-of-distribution SHAMe mock, validating the robustness of our model. This holds down to scales of $k_{\text{max}} = 0.35$ for the galaxy–galaxy power spectrum, and $k_{\text{max}} = 0.25$ for the galaxy–matter cross spectrum and the galaxy bispectrum (see Figs. 6 and 8).
- Increasing the number of bias parameter sets applied to each cosmology achieves a similar precision and accuracy as increasing the number of cosmologies, for the same total number of points, beyond a minimum of around 2,000 cosmologies (see Fig. 11). This could be a very helpful result, as augmenting with more galaxy bias combinations is far faster than generating mocks at new cosmologies.
- For $P_{gg}(k)$ alone, around 20,000 training mocks are required for inference to converge and information content to saturate; for the full statistics combination $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$, the precision continues to increase even with 320,000 training points, indicating that additional mocks (or improved SBI procedures) are needed to saturate the information in those statistics (see Fig. 10).

The *muchisimocks* suite will be publicly available, and is useful for various analyses exploring the information content in galaxy clustering. With its large number of mocks and robust method for modeling the galaxy–dark matter connection, it is ideal for testing approaches to simulation-based inference. The approach could be extended to generate mocks exploring a range of science questions and targeted for specific tracer populations and surveys, paving the way for robust and flexible inference from galaxy clustering.

ACKNOWLEDGEMENTS

The authors would like to thank Beatriz Tucci, Ivana Nikolac, Matt Ho, and Jed Homer; Matteo Zennaro, Daniel López-Cano, Fran-

cisco Maion and the rest of the BACCO group; and Risa Wechsler and the KIPAC Galaxy Formation and Cosmology group for helpful discussions.

SOFTWARE AND LLM USAGE

The following software were used in this work: [ksf: fill in software](#). In this work, large language models (LLMs) were used to make code modifications, write basic code such as plotting functions, support background research, and provide basic L^AT_EX support. Core code and all writing did not involve the use of LLMs.

DATA AVAILABILITY

The *muchisimocks* suite and the code used in this paper will be made publicly available at [thisurl](#).

REFERENCES

- Adame A. G., et al., 2025, *Journal of Cosmology and Astroparticle Physics*, 2025, 008
- Alam S., et al., 2017, *Monthly Notices of the Royal Astronomical Society*, 470, 2617
- Alvey J., Contaldi C. R., Pieroni M., 2026, Simulation-Based Inference with Deep Ensembles: Evaluating Calibration Uncertainty and Detecting Model Misspecification ([arXiv:2507.13495](#)), doi:10.48550/arXiv.2507.13495
- Angulo R. E., Zennaro M., Contreras S., Aricò G., Pellejero-Ibañez M., Stücker J., 2021, *Monthly Notices of the Royal Astronomical Society*, 507, 5869
- Bairagi A., Wandelt B., Villaescusa-Navarro F., 2025, *Astronomy & Astrophysics*, 703, A301
- Bernardeau F., Colombi S., Gaztanaga E., Scoccimarro R., 2002, *Physics Reports*, 367, 1
- Conroy C., Wechsler R. H., Kravtsov A. V., 2006, *The Astrophysical Journal*, 647, 201
- Contreras S., Angulo R., Zennaro M., 2021, *Monthly Notices of the Royal Astronomical Society*, 508, 175
- Contreras S., Angulo R. E., Chaves-Montero J., Kugel R., Schaller M., Schaye J., 2024, *Astronomy & Astrophysics*, 690, A311
- DESI Collaboration et al., 2023, Validation of the Scientific Program for the Dark Energy Spectroscopic Instrument, doi:10.48550/arXiv.2306.06307
- Dawson K. S., et al., 2016, *The Astronomical Journal*, 151, 44
- Desjacques V., Jeong D., Schmidt F., 2017, Technical report, Large-Scale Galaxy Bias. ([arXiv:1611.09787v3](#))
- Doerer L., Jamieson D., Stopyra S., Lavaux G., Leclercq F., Jasche J., 2024, *Monthly Notices of the Royal Astronomical Society*, 535, 1258
- Durkan C., Bekasov A., Murray I., Papamakarios G., 2019, Neural Spline Flows ([arXiv:1906.04032](#)), doi:10.48550/arXiv.1906.04032
- Durkan C., Murray I., Papamakarios G., 2020, in Proceedings of the 37th International Conference on Machine Learning. PMLR, pp 2771–2781
- Euclid Collaboration et al., 2025, *Astronomy & Astrophysics*, 697, A1
- Greenberg D., Nonnenmacher M., Macke J., 2019, in Proceedings of the 36th International Conference on Machine Learning. PMLR, pp 2404–2414
- Hadzhiyska B., García-García C., Alonso D., Nicola A., Slosar A., 2021, *Journal of Cosmology and Astroparticle Physics*, 2021, 020
- Hahn C., Beutler F., Sinha M., Berlind A., Ho S., Hogg D. W., 2018, Likelihood Non-Gaussianity in Large-Scale Structure Analyses, <https://arxiv.org/abs/1803.06348v1>, doi:10.1093/mnras/stz558
- Hahn C., et al., 2022, $\mathcal{S}(\text{IM})\text{BIG}$: A Forward Modeling Approach To Analyzing Galaxy Clustering ([arXiv:2211.00723](#)), doi:10.48550/arXiv.2211.00723
- Hahn C., et al., 2023a, $\mathcal{S}(\text{IM})\text{BIG}$: The First Cosmological Constraints from the Non-Linear Galaxy Bispectrum ([arXiv:2310.15243](#)), doi:10.48550/arXiv.2310.15243
- Hahn C., et al., 2023b, $\mathcal{S}(\text{IM})\text{BIG}$: The First Cosmological Constraints from Non-Gaussian and Non-Linear Galaxy Clustering ([arXiv:2310.15246](#)), doi:10.48550/arXiv.2310.15246
- Hahn C., et al., 2023c, *Journal of Cosmology and Astroparticle Physics*, 2023, 010
- Henriques B. M. B., White S. D. M., Thomas P. A., Angulo R., Guo Q., Lemson G., Springel V., Overzier R., 2015, *Monthly Notices of the Royal Astronomical Society*, 451, 2663
- Jacobus C., Tucci B., Philcox O., 2026, Emulating Cosmic Structure Formation with a Lagrangian Neural Cellular Automaton, <https://arxiv.org/abs/2607.27320v1>
- Jamieson D., Li Y., de Oliveira R. A., Villaescusa-Navarro F., Ho S., Spergel D. N., 2023, *The Astrophysical Journal*, 952, 145
- Jamieson D., Li Y., Villaescusa-Navarro F., Ho S., Spergel D. N., 2024, Field-Level Emulation of Cosmic Structure Formation with Cosmology and Redshift Dependence ([arXiv:2408.07699](#)), doi:10.48550/arXiv.2408.07699
- Kang Y., Bin H., Li D., Hamann J., 2026, TUNeS: Neural Emulation of Large-Scale Structure Across Redshifts ([arXiv:2603.18165](#)), doi:10.48550/arXiv.2603.18165
- Karras T., Laine S., Aittala M., Hellsten J., Lehtinen J., Aila T., 2020, Analyzing and Improving the Image Quality of StyleGAN ([arXiv:1912.04958](#)), doi:10.48550/arXiv.1912.04958
- Kokron N., DeRose J., Chen S.-F., White M., Wechsler R. H., 2021, *Monthly Notices of the Royal Astronomical Society*, 505, 1422
- Lemos P., et al., 2023, SimBIG: Field-level Simulation-Based Inference of Galaxy Clustering ([arXiv:2310.15256](#)), doi:10.48550/arXiv.2310.15256
- Massara E., et al., 2024, SimBIG : Cosmological Constraints Using Simulation-Based Inference of Galaxy Clustering with Marked Power Spectra ([arXiv:2404.04228](#)), doi:10.48550/arXiv.2404.04228
- Modi C., Chen S.-F., White M., 2020, *Monthly Notices of the Royal Astronomical Society*, 492, 5754
- Nguyen N.-M., Schmidt F., Tucci B., Reinecke M., Kostić A., 2024, *Physical Review Letters*, 133, 221006
- Novell-Masot S., et al., 2025, Full-Shape Analysis of the Power Spectrum and Bispectrum of DESI DR1 LRG and QSO Samples ([arXiv:2503.09714](#)), doi:10.48550/arXiv.2503.09714
- Papamakarios G., Murray I., 2016, in Advances in Neural Information Processing Systems. Curran Associates, Inc.
- Peacock J. A., Smith R. E., 2000, *Monthly Notices of the Royal Astronomical Society*, 318, 1144
- Pellejero-Ibañez M., et al., 2026, Validation of the Hybrid Bias Expansion Model for the Galaxy Bispectrum ([arXiv:2606.19452](#)), doi:10.48550/arXiv.2606.19452
- Pellejero-Ibanez M., Stuecker J., Angulo R. E., Zennaro M., Contreras S., Arico G., 2022, *Monthly Notices of the Royal Astronomical Society*, 514, 3993
- Pellejero Ibanez M., Angulo R. E., Jamieson D., Li Y., 2023a, Hybrid-Bias and Displacement Emulators for Field-Level Modelling of Galaxy Clustering in Real and Redshift Space ([arXiv:2307.09134](#))
- Pellejero-Ibanez M., Angulo R. E., Zennaro M., Stuecker J., Contreras S., Arico G., Maion F., 2023b, *Monthly Notices of the Royal Astronomical Society*, 520, 3725
- Philcox O. H. E., Flöss T., 2024, PolyBin3D: A Suite of Optimal and Efficient Power Spectrum and Bispectrum Estimators for Large-Scale Structure, <https://arxiv.org/abs/2404.07249v3>
- Philcox O. H. E., Ivanov M. M., 2022, *Physical Review D*, 105, 043517
- Planck Collaboration et al., 2014, *Astronomy & Astrophysics*, 571, A16
- Qian P. Z. G., 2009, *Biometrika*, 96, 957
- Rubira H., Schmidt F., 2025, Non-Gaussian Galaxy Stochasticity and the Noise-Field Formulation ([arXiv:2511.05484](#)), doi:10.48550/arXiv.2511.05484
- Saadeh D., Koyama K., Morice-Atkinson X., 2025, *Monthly Notices of the Royal Astronomical Society*, 537, 448
- Schaye J., et al., 2015, *Monthly Notices of the Royal Astronomical Society*, 446, 521
- Scoggins M. T., Ho M., Villaescusa-Navarro F., Jamieson D., Doerer L., Bryan G. L., 2025, Learning the Universe: $3\text{H}^{-1}\text{Gpc}$ Tests of a Field Level N -Body Simulation Emulator ([arXiv:2502.13242](#)), doi:10.48550/arXiv.2502.13242
- Seljak U., 2000, *Monthly Notices of the Royal Astronomical Society*, 318, 203
- Shankar F., Lapi A., Salucci P., Zotti G. D., Danese L., 2006, *The Astrophysical Journal*, 643, 14
- Somerville R. S., Davé R., 2015, *Annual Review of Astronomy and Astrophysics*, 53, 51
- Somerville R. S., Hopkins P. F., Cox T. J., Robertson B. E., Hernquist L., 2008, *Monthly Notices of the Royal Astronomical Society*, 391, 481
- Spergel D., et al., 2015, Wide-Field Infrared Survey Telescope-Astrophysics Focused Telescope Assets WFIRST-AFTA 2015 Report ([arXiv:1503.03757](#)), doi:10.48550/arXiv.1503.03757
- Springel V., White S. D. M., Tormen G., Kauffmann G., 2001, *Monthly Notices of the Royal Astronomical Society*, 328, 726
- Springel V., Matteo T. D., Hernquist L., 2005, *Monthly Notices of the Royal Astronomical Society*, 361, 776
- Stevens A. R. H., Croton D. J., Mutch S. J., 2016, *Monthly Notices of the Royal Astronomical Society*, 461, 859

- Stücker J., Pellejero-Ibáñez M., Voivodic R., Angulo R. E., 2025a, *Astronomy & Astrophysics*, 694, A29
- Stücker J., Pellejero-Ibáñez M., Angulo R. E., Maion F., Voivodic R., 2025b, *Astronomy & Astrophysics*, 699, A197
- Tejedor-Cantero A., Boelts J., Deistler M., Lueckmann J.-M., Durkan C., Gonçalves P. J., Greenberg D. S., Macke J. H., 2020, SBI – A Toolkit for Simulation-Based Inference (arXiv:2007.09114), doi:10.48550/arXiv.2007.09114
- Tucci B., Schmidt F., 2024, *Journal of Cosmology and Astroparticle Physics*, 2024, 063
- Ullmo M., Aghanim N., Decelle A., Aragon-Calvo M., 2026, *Astronomy & Astrophysics*, 706, A124
- Vale A., Ostriker J. P., 2004, *Monthly Notices of the Royal Astronomical Society*, 353, 189
- Villaescusa-Navarro F., et al., 2020, *The Astrophysical Journal Supplement Series*, 250, 2
- Villaescusa-Navarro F., et al., 2021, *The Astrophysical Journal*, 915, 71
- Vogelsberger M., et al., 2014, *Monthly Notices of the Royal Astronomical Society*, 444, 1518
- Wang Y., et al., 2022, *The Astrophysical Journal*, 928, 1
- Wang H., Alvarez M., Cooray S., Hearin A., Mansfield P., Storey-Fisher K., Wechsler R. H., 2026, Stress-Testing Cosmological Inference on Small Scales with SHAM and HOD Galaxy–Halo Connection Models, in preparation
- Wechsler R. H., Tinker J. L., 2018, *Annual Review of Astronomy and Astrophysics*, 56, 1
- York D. G., et al., 2000, *The Astronomical Journal*, 120, 1579
- Zennaro M., Angulo R. E., Contreras S., Pellejero-Ibáñez M., Maion F., 2022, *Monthly Notices of the Royal Astronomical Society*, 514, 5443
- Zennaro M., Angulo R. E., Pellejero-Ibáñez M., Stücker J., Contreras S., Aricò G., 2023, *Monthly Notices of the Royal Astronomical Society*, 524, 2407
- Zheng Z., et al., 2005, *The Astrophysical Journal*, 633, 791
- de Santi N. S. M., et al., 2023, *The Astrophysical Journal*, 952, 69

APPENDIX A: VALIDATION OF INFERENCE ON MEAN DATA VECTOR

In this analysis, we have shown results of the SBI pipeline with the *muchisimocks* forward model on the “cosmic variance” test set, which contains 1,000 realizations run with different seeds at the same cosmology, bias, and noise parameters. Specifically, we took the mean of the 1,000 data vectors computed on each of these mocks individually, and used that mean data vector as the input to the SBI model. In this section, we investigate this analysis choice and its implications.

We first compute the statistics on all 1,000 mock realizations, and perform the inference on each individually with our SBI pipeline. We obtain a posterior for each of the 1,000 realizations. To visualize the “typical” posterior, we compute the mean-of-the-means of the 1,000 posteriors, and shift each posterior such that it is centered on that value; we then pool the posteriors, drawing 500 samples from each.

We then compute the mean of the statistics of the 1,000 mocks, and perform the inference on that mean data vector. This mean data vector has the benefit of reducing the effect of cosmic variance on the statistic, but brings the complication of having a different covariance than the data the SBI model was trained on (namely, much lower variance). In standard MCMC approaches, this would be handled by appropriately scaling the covariance matrix, but in the context of SBI, the covariance is learned by the model from the training data and is not given explicitly. Thus it is important to understand the behavior of the model when we use the mean data vector for inference.

Figure A1 shows the result of these ways of analyzing the cosmic

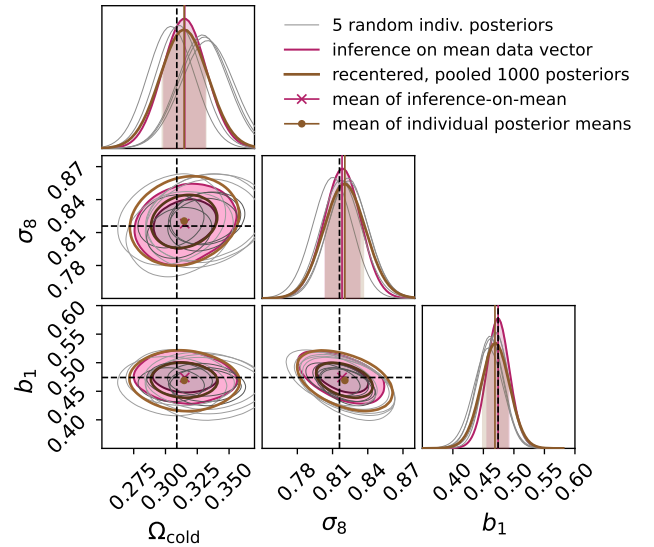


Figure A1. Validation of inference on the mean of the data vectors for the cosmic variance test set, for the full $P_{gg}(k) + B_{ggg}(k_1, k_2, k_3) + P_{gm}(k)$ combination. The grey contours show posteriors from inference performed on statistics of 5 randomly selected realizations from the test set; the brown contours are pooled samples from each of the 1,000 realizations, 500 of each recentered to the mean of the means; the magenta contours show inference performed on the mean of 1,000 data vectors. The mean of the inference-on-the-mean (magenta vertical line and X marker) nearly exactly agrees with the mean of the 1,000 individual posterior means (brown vertical line and dot marker). The shape and width of the posteriors are also broadly consistent.

variance test set, all for the combined statistic $P_{gg}(k) + P_{gm}(k) + B_{ggg}(k_1, k_2, k_3)$. The grey contours show posteriors of 10 randomly selected individual mocks; the brown contour shows the recentered-pooled posteriors (over all 1,000 mocks); and the magenta contour shows the inference performed on the mean of the 1,000 data vectors. The brown dot and vertical lines show the mean of the means of the 1,000 individual posteriors.

We see that the inference on the mean produces a posterior very close to that of the recentered-pooled posterior. The means, as well as the MAPs, for these two results are within $\sim 1\%$ for all parameters; both are within 0.5σ (2%) from the true values. The predicted uncertainty regions are also very similar: the symmetrized inner 68% width agrees to 0.2% for Ω_m , 4% for σ_8 , and 4% for b_1 . We check that the recentered-pooled posterior width agrees with the mean of the widths of the 1,000 individual posteriors, to better than 0.5%. The inference-on-the-mean does generally produce slightly tighter constraints than the average of the individual posteriors, interpreted as mild overconfidence. However, the mean data vector should be less affected by prior effects than individual realizations with summary statistics towards the edges of the distribution. *ksf: i think we still don't know exactly how to interpret this; so i am not sure how to tie these last couple sentences together.*

This analysis indicates that despite the lower variance of the mean data vector, the SBI pipeline produces a posterior that has the typical width of a posterior estimated with the noisier data vector, as that was closer to the noise levels of data it was trained on. We thus use the inference on the mean data vector throughout the paper to be able to show the typical result of the inference, with the effect of cosmic variance (which would cause a bias in any given individual sample) removed. We emphasize that there is not a clear statistical interpre-

tation of this inference-on-the-mean for simulation-based inference, and the stability of the comparison to the individual posteriors varies somewhat across models. However, the results empirically demonstrate that this is a useful way to validate an SBI pipeline with a much lower computational burden compared to sampling many individual posteriors.

APPENDIX B: FULL POSTERIOR CONTOURS

We show the full contour plots for all statistics combinations considered, for all of the varied parameters: the two cosmological parameters, four bias parameters, and five noise parameters. Figure B1 shows the posteriors inferred from the mean of the data vectors of the 1,000-mock cosmic variance test set. Figure 6 shows the inference on the out-of-distribution SHAMe mock at fiducial number density, $\bar{n} = 2.2 \times 10^{-4} (h^{-1}\text{Mpc})^{-3}$.

This paper has been typeset from a $\text{\LaTeX}$ file prepared by the author.

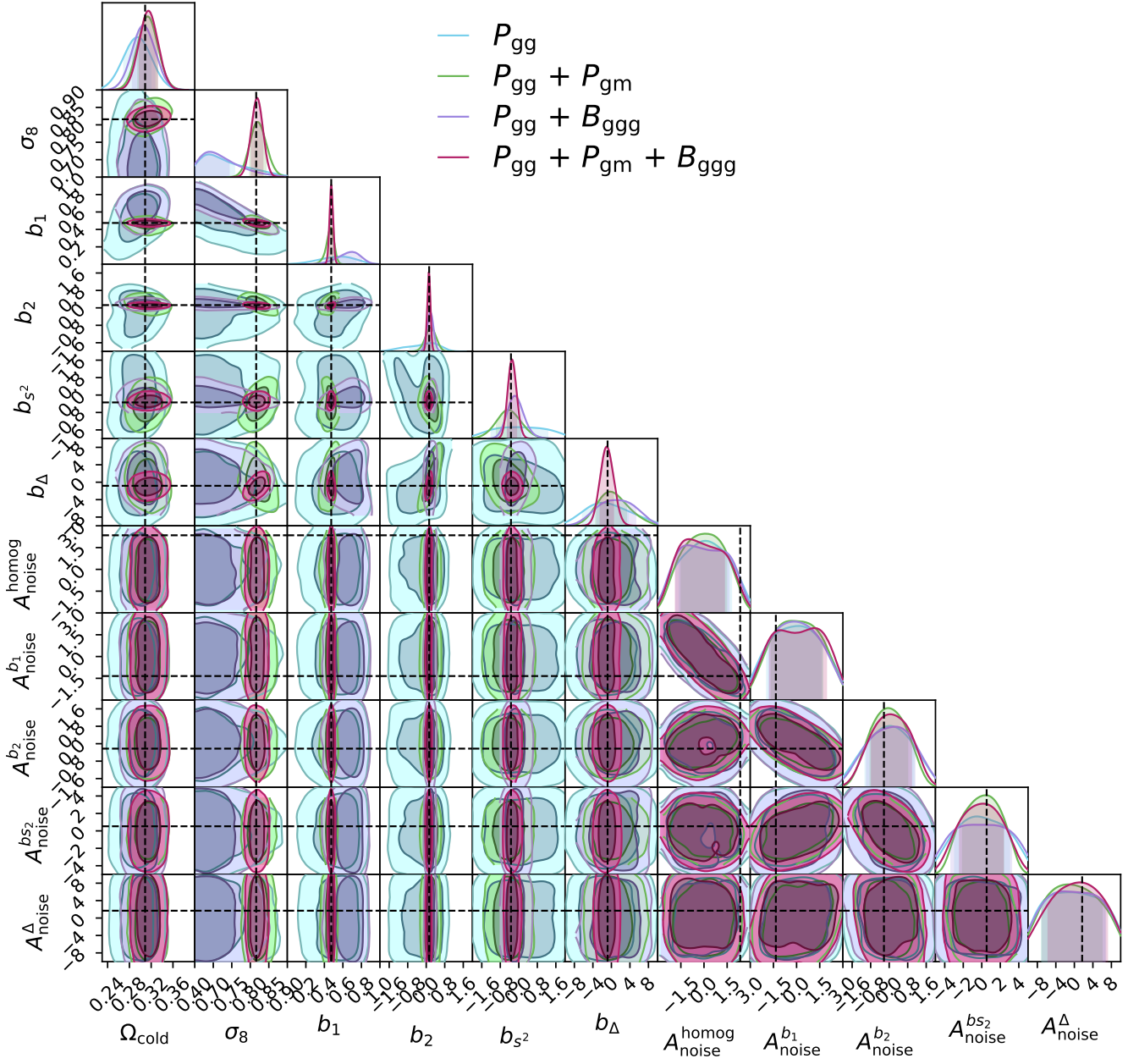


Figure B1. Full posterior contours for the cosmic variance test set, inferred on the mean of the data vectors across this set.

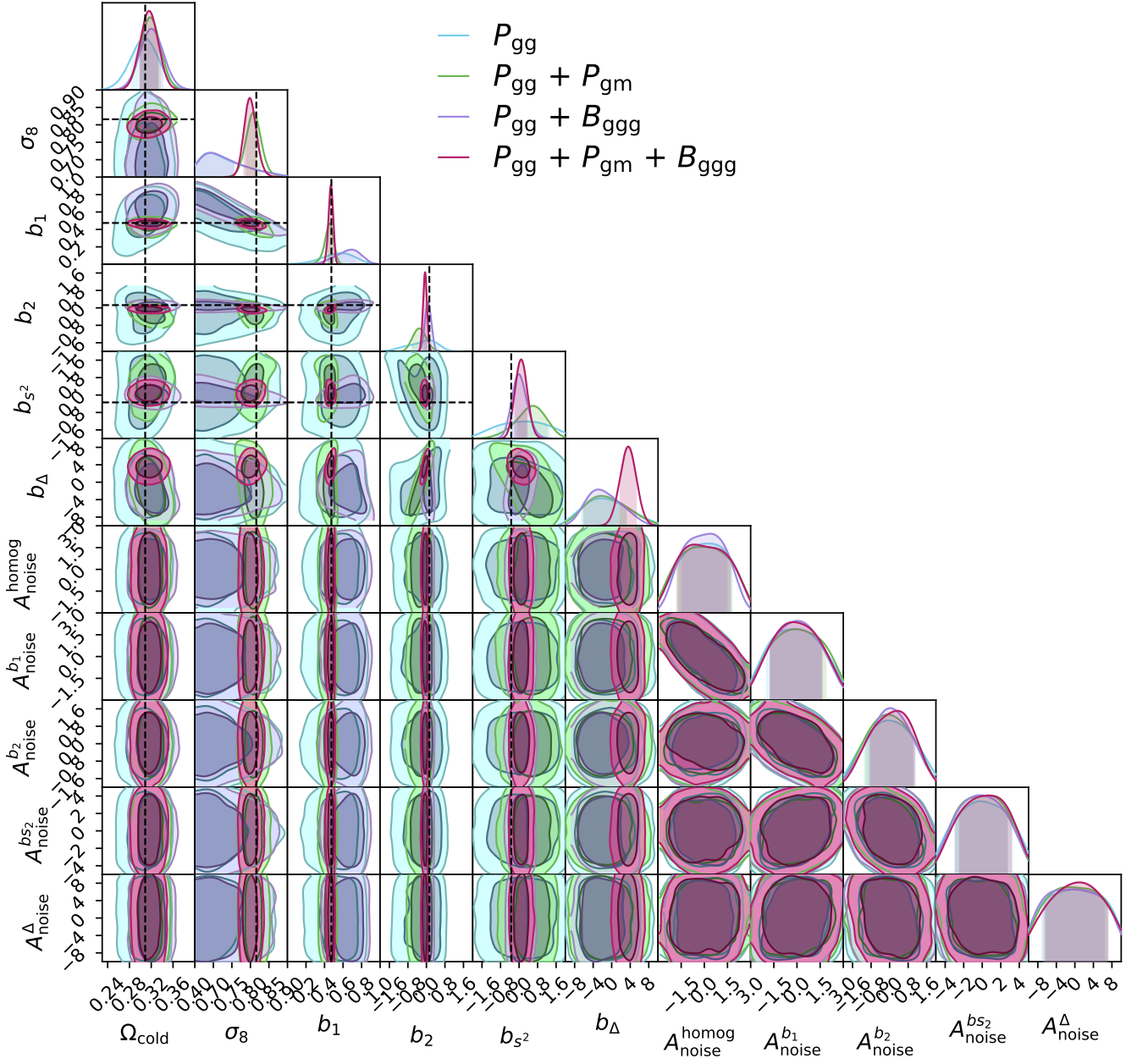


Figure B2. Full posterior contours for the SHAME out-of-distribution test set with $\bar{n} = 2.2 \times 10^{-4} (h^{-1} \text{Mpc})^{-3}$.